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Abstract

Deploying Vision-Language Models (VLMs) on autonomous mobile robots is severely constrained by the strict computational, memory, and thermal limits of edge hardware. While fine-tuning or retraining foundation models for adverse operational environments is impractical on embedded platforms, compact edge decoders frequently suffer from spatial grounding deficits, resolution downsampling loss, and temporal redundancy in continuous video streams. This thesis investigates an inference-time visual preprocessing architecture designed to enhance VLM scene comprehension and downstream reasoning without modifying underlying model weights.

We propose and evaluate four task-aware spatial prompting techniques Context-Aware Zoom (CAZ), Query-Aware Bounding Box (QABB), Semantic Hazard Isolation (SHI), and Perception Metric Grounding (PMG) alongside dual-track query-aware video keyframe extraction pipelines (QA-IF and Smart Sampling). To systematically benchmark these strategies across the cognitive hierarchy of mobile robotics, we curated a benchmark corpus spanning seven functional task domains, comprising 2,103 static QA pairs across 731 visual scenes (with a 30% low-light stress test) and 70 continuous robotic video trajectories annotated with minimal keyframe triggers. Real-time tensor inference and telemetry profiling were benchmarked on an NVIDIA Jetson AGX Thor across three representative model tiers: LLaVA-Phi3 (3.8B), Qwen3-VL (4B), and Gemma-4 (12B). To overcome the blind spots of conventional lexical metrics, deep semantic correctness was evaluated via an automated, three-round heterogeneous multi-agent debate jury (DeepSeek-R1, Qwen2.5, and Llama-3) offloaded to an HPC cluster.

Across 12,250 multimodal benchmark instances, our empirical findings reveal that inference-time adaptation fundamentally redefines edge vision-language perception through two governing trade-offs: attentional focus versus global context in static imagery, and extraction latency versus token processing economy in continuous video. While targeted spatial prompts like bounding overlays and zoom crops improve fine-grained object localization by up to +8.81%, they impose up to a -19.53% penalty on open-ended scene understanding by stripping peripheral context—demonstrating that visual prompts must be applied conditionally rather than globally. Under adverse sensing conditions, lightweight photometric regularizers such as Gamma Correction reliably stabilize low-light hazard detection without model retraining. In dynamic video streams, while traditional shot detectors fail catastrophically (94.29% error rate) under smooth robot camera motion, native container demuxing achieves zero-drop real-time throughput (0.45s latency), and query-conditioned frame pruning compresses visual payloads down to 1.74 frames to prevent downstream edge VLM token saturation. Ultimately, our analysis uncovers a structural decoupling where standard NLP metrics reward linguistic fluency while overlooking physical hallucinations, establishing that task-conditioned visual preprocessing paired with grounded multi-agent factual auditing provides a practical, robust foundation for embodied edge autonomy.

die Zusammenfassung

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Preface

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1 Introduction

Reliable scene comprehension is a fundamental requirement for autonomous robotic systems operating in dynamic, unstructured, and safety-critical environments. The ability of robots to perceive and interpret their surroundings in real time directly impacts navigation, decision-making, and interaction with humans or other agents. Vision Language Models (VLMs), supported by Large Language Models (LLMs), provide powerful multimodal reasoning capabilities, allowing robots to understand visual scenes, recognize relevant entities, and generate context-aware explanations or instructions.

Despite their capabilities, the performance of VLMs on edge devices remains highly sensitive to the quality and variability of incoming visual data. Changes in illumination, motion blur, partial occlusions, sensor noise, and unconventional viewpoints can significantly degrade model outputs, compromising robustness and safety in real-world deployments. Traditional methods for improving scene comprehension, such as expanding training datasets or fine-tuning models for specific environments, are often impractical in edge robotics. These approaches require substantial computational resources, large annotated datasets, and repeated training cycles, and they cannot address the moment-to-moment fluctuations that occur during live perception.

To overcome these limitations, there is a growing interest in inference-time, on-device image augmentation and transformation techniques. Such methods preprocess visual inputs before they are fed into VLMs, enhancing robustness without altering model parameters. Examples of these transformations include illumination normalization, geometric adjustments, highlighting structural features, and generating alternative perspectives. By acting as an adaptive preprocessing layer, these techniques provide VLMs with more interpretable and stable representations of the scene, even under imperfect sensing conditions.

Recent advances in lightweight neural architectures and edge-computing hardware make this approach increasingly feasible. While LLMs contribute high-level reasoning and context fusion, VLMs enable real-time understanding of camera streams, sensor imagery, and scene-level semantics. Integrating lightweight, computationally efficient augmentation methods at inference time can therefore strengthen perception on edge platforms without the need for additional training.

This thesis investigates a software architecture for edge-based image augmentation and transformation aimed at improving VLM/LLM scene comprehension in autonomous robotic systems. The proposed system ingests live visual data, applies real-time transformations, and interfaces with multimodal inference models deployed on robotic platforms. The primary objectives include enhancing scene interpretation, generating structured descriptions, and providing context-aware assessments usable for downstream autonomy tasks such as navigation, human-robot interaction, and safety monitoring. Special emphasis is placed on **real-time feasibility, robustness under variable sensing conditions, and the integration of augmentation methods into edge inference pipelines** without reliance on model retraining or fine-tuning.

2 Literature Review

This chapter reviews research related to inference-time augmentation and transformation techniques that can enhance scene understanding in VLMs. Since robotic perception systems frequently rely on both static images and continuous video streams, the literature review is divided into two main categories: methods designed for video inputs and methods applied to individual images. The goal is to identify preprocessing strategies that can improve model robustness and interpretability without requiring retraining or modification of the underlying models.

2.1 Image-Based Augmentation and Transformation Methods

Image-based augmentation methods modify individual visual samples before they are processed by a model. These transformations aim to improve visual clarity, normalize viewpoint variations,

or induce controlled perturbations under adverse sensing conditions (Kumar et al., 2024). Such inference-time preprocessing strategies are particularly critical for edge-deployed robotic systems, where strict computational and real-time bounds preclude extensive parameter fine-tuning or online retraining.

Common inference-time image transformations encompass geometric modifications, photometric adjustments, structural annotations, and synthetic domain transformations. Geometric transformations including image flipping, rotation, scaling, random cropping, shear mapping, elastic deformation, and perspective warping normalize viewpoint shifts and maintain spatial invariance. Photometric methods, such as color jittering, solarization, kernel blurring or sharpening, and synthetic noise injection, aim to improve visual clarity under challenging illumination or sensor noise conditions (Kumar et al., 2024; Shorten and Khoshgoftaar, 2019).

To prevent models from over-relying on local textures, structural and compositional augmentations alter spatial information density. Explicit annotation overlays (e.g., bounding box boundaries and segmentation masks) provide non-destructive spatial prompts, whereas region-dropping strategies such as Random Erasing (Zhong et al., 2020) and GridMask (P. Chen et al., 2024) selectively occlude contiguous or structured image regions. Furthermore, environmental domain-shift operations (e.g., weather/fog simulations and style transfer filters) alongside Test-Time Augmentation (TTA) ensembles evaluate model prediction stability across diverse perturbation spaces (Shanmugam et al., 2021).

The following sections review selected image-based augmentation techniques spanning segmentation, depth estimation, saliency guidance, bounding box annotations, environmental enhancement, and super-resolution and discuss their potential to improve scene comprehension in VLM-based perception systems.

2.1.1 Segmentation-Based Approaches

Segmentation provides pixel-wise environmental understanding, enabling the precise delineation of complex object boundaries required for autonomous navigation and visual grounding. The visual segmentation domain was fundamentally redefined by the Segment Anything Model (SAM), which formulated class-agnostic mask generation as a promptable foundation task (Kirillov et al., 2023). SAM leverages a heavy Vision Transformer (ViT-H) image encoder pre-trained via Masked Autoencoders to extract high-dimensional visual embeddings, which are cross-attended with point, box, or text prompts in a lightweight decoder head. To resolve semantic ambiguities where a single prompt maps to multiple valid object boundaries, the decoder predicts multi-mask candidates alongside associated Intersection-over-Union (IoU) confidence scores (Kirillov et al., 2023). However, the heavy ViT backbone introduces substantial latency constraints requiring roughly 456ms per frame rendering vanilla SAM impractical for real-time edge perception on mobile platforms (X. Zhao et al., 2023).

To bridge this latency gap, recent research has focused on lightweight Convolutional Neural Network (CNN)-based instance segmentation and model distillation. Real-time architectures such as the YOLOv8 segmentation pipeline (Jocher, Chaurasia and Jing, 2023) and FastSAM (X. Zhao et al., 2023) reformulate promptable segmentation as a single-pass instance segmentation task. By replacing heavy ViT encoders with Cross-Stage Partial (CSP) Darknet backbones, these models compute prototype masks and geometric coefficients in a single forward pass, achieving inference speeds up to 50 times faster than SAM while preserving exact spatial boundaries. Concurrently, MobileSAM optimizes the transformer pipeline via a Decoupled Distillation framework (C. Zhang et al., 2023). By training a compact “Tiny-ViT” student network to mimic the high-dimensional feature representation of the ViT-H teacher, MobileSAM compresses the encoder parameter footprint from 632M to 9.66M, enabling <15ms inference speeds on mobile edge devices without sacrificing mask fidelity.

Beyond foundation models, specialized geometric and foreground segmentation frameworks serve as vital spatial pre-filters for VLMs. Background removal tools such as Rembg utilize nested U-structures (U²-Net) to extract high-resolution foreground objects and eliminate high-entropy visual

clutter (Qin et al., 2020). Furthermore, hybrid vision-language pipelines combine open-vocabulary detectors with promptable segmenters such as Grounded-SAM (Ren et al., 2024) and CLIPSeg (Lüddecke and Ecker, 2022) to translate abstract natural language queries directly into contoured visual masks. In resource-constrained perception pipelines, lightweight geometric segmentation layers isolate target entities before passing condensed visual inputs to downstream foundation models.

2.1.2 Depth and Geometry Models-Based Approaches

Monocular depth estimation translates two-dimensional visual inputs into high-dimensional spatial topographic maps, directly addressing the spatial reasoning and visual-grounding deficits prevalent in standard VLMs (B. Chen et al., 2024). Foundational work in monocular depth, such as the MiDaS framework, established zero-shot spatial generalization by training on multi-dataset mixes to resolve scale and shift ambiguities inherent to disparate depth sensors (Ranftl et al., 2020). MiDaS excels at relative depth estimation, effectively separating foreground entities from background clutter; however, its scale-invariant predictions lack the absolute physical scale necessary for fine-grained robotic manipulation and precise path planning (Bhat et al., 2023).

To bridge the gap between relative spatial ordering and absolute physical distance, ZoeDepth introduced a hybrid metric-relative depth framework (Bhat et al., 2023). Built upon Dense Prediction Transformers, ZoeDepth addresses the scale-invariant nature of relative models by incorporating a metric bins module. This module discretizes the continuous depth spectrum into N_{bin} discrete bins and predicts bin probabilities to output physical distances in meters:

$$d_{\text{metric}} = \sum_{i=1}^{N_{\text{bin}}} p_i \cdot b_i \quad (1)$$

where p_i represents the predicted probability distribution and b_i denotes the center value of the i -th metric bin (Bhat et al., 2023). More recently, foundation depth models such as Depth Anything have scaled monocular depth estimation by leveraging massive unlabeled video corpora through self-supervised pre-training and feature distillation (Yang et al., 2024). Depth Anything demonstrates unprecedented robustness against visual degradation, specular reflections, and motion blur, making it a powerful spatial feature extractor for edge-deployed perception pipelines.

Beyond raw depth maps, geometric topology can be extracted through 3D surface normal estimation (Horn, 1986). By computing spatial depth gradients across monocular depth tensors, surface normal vectors capture micro-geometric orientations independently of surface texture or lighting fluctuations (Bae, Budvytis and Cipolla, 2021). When color-coded and rendered as visual overlays, surface normal maps provide explicit planar and curvature cues, allowing VLMs to distinguish between navigable ground planes, obstacles, and structural boundaries without requiring weight fine-tuning.

2.1.3 Saliency and Attention Guidance Segmentation

Visual saliency filters spatial attention, directing model focus toward task-relevant visual regions while stripping out uninformative background clutter. Traditional deep learning architectures such as U^2 -Net utilize nested U-structures with Receptive Field Blocks (RSU) to capture multi-scale spatial details and global context simultaneously (Qin et al., 2020). By synthesizing side-output saliency maps across multiple resolutions, U^2 -Net isolates class-agnostic salient objects without requiring explicit bounding box supervision. When mapped via jet colormaps or binary threshold masks, these saliency maps enable dynamic cropping workflows that preserve fine-grained structural features while eliminating high-entropy environmental noise.

While class-agnostic saliency models effectively localize generic visual highlights, robotic agents operating under natural language instructions require task-conditioned attention gating. To bridge text semantics with visual focus, language-conditioned segmentation models like CLIPSeg extend

pre-trained vision-language backbones with compact transformer decoders (Lüddecke and Ecker, 2022). CLIPSeg generates dense spatial activation maps conditioned on targeted text prompts, allowing perception pipelines to isolate specific query-relevant entities in highly cluttered environments (Lüddecke and Ecker, 2022; H. Shi et al., 2022).

However, standard VLM visual encoders aggressively downsample high-resolution visual streams (e.g., to 224×224 pixels), which frequently erases fine text, small indicators, or distant objects (Radford et al., 2021). While language-conditioned attention maps localize regions of interest, failing to rescale or zoom into these localized regions prior to tokenization results in significant visual information loss. Grounding vision-language models on small spatial targets thus requires dynamic, query-aware visual focusing strategies that preserve localized spatial detail before passing visual tokens to downstream foundation models (Lüddecke and Ecker, 2022).

2.1.4 Object Detection and Bounding Box-Based Annotations

Spatial object detection provides structured bounding boxes that anchor downstream VLMs to localized scene regions. Real-time end-to-end detection transformers, such as RT-DETR, eliminate Non-Maximum Suppression (NMS) post-processing latency by utilizing bipartite matching loss functions and IoU-aware query selection mechanisms. By decoupling multi-scale feature interactions through an efficient hybrid CNN-Transformer encoder, RT-DETR achieves high detection accuracy and -10ms inference speeds, providing robust region proposals for resource-constrained edge platforms (Y. Zhao et al., 2024).

While traditional object detectors operate on predefined Common Objects in Context (COCO) taxonomies, open-vocabulary architectures unify vision and language embeddings to enable prompt-driven detection. YOLO-World integrates a Re-parameterizable Vision-Language Path Aggregation Network (RepVL-PAN) to connect multi-scale visual features with text prompt embeddings in real time (Cheng et al., 2024). This allows mobile perception systems to execute dynamic visual grounding for arbitrary open-set categories without requiring model fine-tuning.

Beyond axis-aligned object boundaries, human-robot interaction and structural scene understanding require fine-grained kinematic annotations. Modern YOLO pose estimation pipelines extract high-confidence keypoint coordinates alongside bounding boxes, constructing skeletal topological trees that explicitly represent spatial geometry and body orientation (Jocher, Chaurasia and Jing, 2023). In real-time perception pipelines, these structural overlays and localized bounding boxes reduce spatial ambiguity, providing explicit visual tokens before downstream scene comprehension.

2.1.5 Photometric, Color, and Radiometric Enhancements

Adverse environmental conditions such as low illumination, extreme glare, or non-uniform shadows severely degrade visual signal quality, leading to feature loss in deep vision backbones. Photometric and radiometric enhancement techniques act as deterministic pre-filtering operations to restore contrast and normalize illumination across non-ideal sensing environments. Standard non-linear radiometric adjustments, such as Gamma Correction, dynamically rescale pixel luminance distributions using exponential power-law transformations, enhancing low-light visibility while preventing saturation in high-exposure regions (Shorten and Khoshgoftaar, 2019).

To resolve localized illumination imbalances without amplifying background noise, Contrast Limited Adaptive Histogram Equalization (CLAHE) partitions images into contextual tiles and applies localized histogram equalization bounded by a contrast clip limit (Reza, 2004). CLAHE effectively restores fine textures in underexposed or overexposed regions, making it an essential preprocessing tool for thermal and low-light robotic perception (Shorten and Khoshgoftaar, 2019).

For complex illumination artifacts, Retinex-based algorithms decompose images into reflectance and illumination components based on Land’s Retinex theory (Land, 1977). Single-Scale Retinex (SSR) applies logarithmic Gaussian filtering to isolate invariant object reflectance from dynamic scene illumination (Jobson, Rahman and Woodell, 1997). By removing dynamic lighting vari-

ations while preserving spatial reflectance, SSR ensures consistent color representation and feature stability for downstream foundation models under severe environmental illumination shifts.

2.1.6 Structured Domain and Graph-Based Representations

To operate effectively across unconstrained environments, perception pipelines must combine localized visual features into broader structural and semantic contexts. Composite augmentation strategies, such as Mosaic Augmentation, stitch four distinct training images into a single composite view (Bochkovskiy, C.-Y. Wang and Liao, 2020). By forcing detection models to evaluate objects across four different context frames simultaneously, Mosaic augmentation enriches spatial diversity, artificially increases batch context, and improves small object detection without incurring additional memory overhead (Bochkovskiy, C.-Y. Wang and Liao, 2020).

To enhance robustness against synthetic domain shifts and stylistic variations, neural style transfer filters map source visual inputs onto diverse artistic or environmental textures (Gatys, Ecker and Bethge, 2016). By decoupling content representations from style distributions using deep convolutional features, style-transferred representations prevent downstream models from over-fitting to specific color distributions or lighting conditions, thereby improving out-of-distribution generalization.

Beyond pixel-level transformations, high-level visual reasoning benefits from explicit relational structures. Scene Graph Generation (SGG) converts unstructured visual inputs into grounded directed graphs, where nodes represent localized entities and edges capture spatial or semantic relationships (e.g., “cup *on* table”) (Krishna et al., 2016; Xu et al., 2017). While early SGG models suffered from prohibitive inference latencies, modern real-time architectures such as REACT++ balance relational accuracy and execution speed by deploying parameter-efficient subject-to-object cross-attention within a prototype space. By coupling single-stage object detectors with streamlined cross-modal attention mechanisms and Open Neural Network Exchange (ONNX) tensor runtime acceleration, high-throughput SGG models rapidly extract subject-predicate-object triplets directly from raw visual streams. Overlaying these structured graph dependencies onto visual inputs provides Vision-Language Models with explicit spatial priors, mitigating hallucination and improving contextual reasoning in embodied robotics (Neau and Falomir, 2026).

2.1.7 Comparative Summary of Techniques

The visual transformation strategies reviewed in this section represent a spectrum of pre-processing mechanisms designed to enhance vision-language perception across unconstrained operational environments. As summarized in Table 1, these methodologies range from low-level radiometric adjustments which normalize environmental factors like illumination and noise to complex geometry and segmentation models that inject explicit 3D spatial priors and structural masks directly into the visual input space.

By systematically structuring visual inputs before tokenization, these pre-processing techniques directly address the core vulnerabilities of standard VLMs, including spatial grounding deficits, sensitivity to lighting variations, and resolution downsampling bottlenecks. Table 1 provides a structured comparative synthesis of all established visual transformation techniques analyzed in Section 2.1, categorizing them by category, method, perception level, primary VLM benefit, and computational overhead.

Table 1: Comparative Summary of Established Visual Augmentation and Pre-processing Techniques for Vision-Language Models.

Category	Method	Perception Level	Primary VLM Benefit	Overhead
Basic Taxonomy	Geometric Transformations	Pixel / Global	Teaches shift, rotation, and scale invariance	Low
	Photometric Modifications	Pixel	Restores signal contrast under lighting fluctuations	Low
	Structural Masking	Global	Forces global contextual reasoning via spatial dropout	Low
	Domain Shifts & TTA	Global	Improves out-of-distribution robustness & stability	Medium
Segmentation	YOLOv8-Seg	Object	Provides fast geometric boundary contours	Medium
	FastSAM / MobileSAM	Object	Delivers real-time class-agnostic instance masks	Medium
	Rembg (U ² -Net)	Object / Scene	Strips uninformative background clutter	Medium
Depth & Geometry	MiDaS	Scene	Generates relative 3D spatial depth maps	High
	ZoeDepth	Scene	Predicts metric physical distances in meters	High
	Depth Anything	Scene	High-precision depth under severe visual degradation	High
	Surface Normalization	Scene	Captures 3D planar and surface orientation	High
Saliency	SaliencyCrop	Object	Isolate task-relevant salient objects	Medium
Detection & Pose	RT-DETR	Object	Low-latency end-to-end bounding box proposals	Medium
	YOLO-World	Object	Open-vocabulary promptable visual grounding	Medium
	YOLO Pose Estimation	Object	Skeletal topological keypoint extraction	Medium
Photometric	Gamma Correction	Pixel	Rescales pixel luminance distribution non-linearly	Low
	CLAHE	Pixel	Tile-based localized contrast enhancement	Low
	RetinexSSR	Pixel	Isolates invariant object reflectance from lighting	Medium
Structured Domain	Mosaic Augmentation	Global	Enhances multi-object spatial context density	Low
	Style Transfer Filters	Global	Decouples visual content from domain style shifts	High

2.2 Video-Based Augmentation and Transformation Methods

In the context of processing long-form video data for Edge Vision-Language Models, traditional video augmentation techniques that alter the entire continuous pixel volume across every frame are computationally prohibitive (Brkic et al., 2026). Modern multimodal architectures remain severely constrained by strict context windows, limited hardware resources, and visual token capacities, making the direct ingestion of raw, high-frame-rate streams infeasible for real-time deployment on resource-constrained edge platforms (S. Zhang et al., 2025).

To overcome this bottleneck, this project redefines the paradigm of “video augmentation” by treating it fundamentally as an optimized Frame Extraction and Selection pipeline. Rather than passing the entire raw video asset to the downstream model, the pipeline dynamically condenses the temporal sequence into a highly curated, structurally dense subset of keyframes. Once this critical subset is mathematically or semantically isolated, localized downstream image augmentation and inference-time enhancements can be executed with maximum computational efficiency.

The validity of prioritizing smart frame selection over raw uniform downsampling is heavily supported by recent architectural benchmarks, which demonstrate that unguided frame sampling introduces substantial frame-sampling bias, creates temporal fragmentation, and fills strict model context windows with redundant visual noise or irrelevant data (Brkic et al., 2026; S. Zhang et al., 2025). Standardizing the sampling protocol under a controlled token budget has been shown to drastically shift model performance rankings and unmask true visual representation capacities. Therefore, the core objective of video-level transformation in this framework is to maximize semantic information density per token budget while minimizing the expensive preprocessing and inference time overhead typically associated with complex optimization stages.

To systematically evaluate and benchmark this approach, we implement and analyze distinct frame extraction methodologies across classical, deep clustering, and query-aware paradigms:

2.2.1 Structural and Low-Level Pixel Methods

Low-level pixel and structural extraction methods operate directly on raw visual descriptors such as color space distributions, gradient orientation maps, wavelets, or codec-level container metadata. Modern video compression standards (Moving Picture Experts Group (MPEG-4), Advanced Video Coding (AVC/H.264) and High Efficiency Video Coding (HEVC/H.265)) organize video sequences into a Group of Pictures (GOP) structure comprising three primary frame types: Intra-coded (*I*-frames), Predicted (*P*-frames), and Bi-directionally predicted (*B*-frames) (Sadiq et al., 2020). While *P*-frames and *B*-frames compress video payload by storing only temporal motion vectors and spatial residual error deltas relative to adjacent reference frames, *I*-frames serve as spatially

self-contained, fully decodable visual snapshots. Consequently, at the lowest processing tier, zero-decoding heuristic baselines bypass inter-frame motion reconstruction entirely, extracting native *I*-frames directly from video bitstream headers based on fixed encoder keyframe intervals rather than underlying visual semantics (Sadiq et al., 2020).

To move beyond static codec keyint settings, temporal shot boundary detection algorithms evaluate color variance across successive frames to extract representative shot keyframes while suppressing over-sampling during continuous camera panning (Ciocca and Schettini, 2006; Sadiq et al., 2020). Similarly, statistical thresholding techniques compute frame-to-frame pixel or histogram intensity differences against dynamic thresholds derived from rolling mean and standard deviation metrics to detect transition boundaries, while bitwise XOR operations across binary pixel matrices offer bit-level execution speeds for identifying sudden spatial shifts (Rashmi and Nagendraswamy, 2016; Sadiq et al., 2020).

To capture more nuanced perceptual and visual changes, intermediate pixel-based methods incorporate structural and spectral representations. Perceptual metrics such as the Structural Similarity Index (SSIM) evaluate luminance, contrast, and local structure sequentially, dynamically triggering new keyframes only when visual similarity drops below a specified threshold (Z. Wang et al., 2004). Global visual shifts can also be monitored through joint color histograms evaluated via distance metrics (H. Zhang, Kankanhalli and Smoliar, 1993), or through local frame covariance matrices where structural dissimilarity is identified by minimal eigenvalue states (Benni et al., 2015; Sadiq et al., 2020). Furthermore, multi-descriptor approaches such as the Curvature Points (CP) algorithm combine color histograms, edge direction maps, and wavelet statistics into a cumulative difference curve, using corner detection to extract keyframes at visual inflection midpoints without forcing artificial cluster counts. To ensure visual quality across all these low-level heuristics, Laplacian variance filtering is frequently applied as a spatial sharpness gate to prune blurry or corrupted frames resulting from rapid camera movements (Ciocca and Schettini, 2006).

2.2.2 Motion and Kinematic-Based Methods

While histogram-based approaches analyze global appearance changes between consecutive frames, they remain insensitive to the underlying motion dynamics responsible for those changes. To address this limitation, motion-aware frame extraction techniques employ optical flow analysis to estimate the physical displacement of visual features across time, isolating keyframes during critical motion phase shifts such as the onset, peak, or deceleration of camera panning and object trajectories (Sadiq et al., 2020).

Classical sparse optical flow formulations first identify stable corner features using the Shi-Tomasi criterion and subsequently track their motion vectors between adjacent frames using the Lucas-Kanade optical flow algorithm (Lucas and Kanade, 1981; J. Shi and Tomasi, 2000). Evaluating the average magnitude and acceleration of these sparse feature vectors provides a proxy for visual dynamics, enabling dynamic frame extraction frameworks such as AdaFrame to select keyframes at temporal motion acceleration peaks (Zuxuan Wu et al., 2019). For complex or unconstrained environments involving simultaneous camera motion and multi-object movement, dense optical flow techniques (e.g., Farneback flow expansion) compute continuous motion velocity fields across the entire pixel grid (Farneback, 2003). Frameworks like Motion-Guided Samplers (MGSampler) integrate this spatial flow energy over time to quantify global kinetic activity, selecting keyframes at kinetic inflection points to preserve complex motion transitions and object manipulations with high temporal fidelity (S. Zhang et al., 2025; Zhi et al., 2021).

2.2.3 Unsupervised Deep Latent and Graph-Based Clustering

Unsupervised feature-space partitioning methods project raw video frames into high-dimensional latent representations, grouping visual samples based on semantic feature proximity rather than raw pixel differences (Kuanar, Panda and Chowdhury, 2013; Tang et al., 2022). Classical color-space partitioning frameworks convert frames into perceptual color spaces (e.g., LUV) and perform *K*-Means clustering alongside spatial blur filtering to reject uninformative or out-of-focus visual

samples (Sadiq et al., 2020). Modern deep latent variants replace low-level color histograms with deep visual embeddings extracted from pre-trained Vision Transformers or CNN backbones (e.g., Contrastive Language-Image Pre-training (CLIP) image encoders). Executing K -Means medoid clustering directly on these deep embeddings isolates cluster centroids that correspond to distinct semantic concepts (Tang et al., 2022).

To avoid the uniform variance assumptions inherent to standard K -Means, advanced algebraic and geometric partitioning methods optimize feature subspace coverage (Kuanar, Panda and Chowdhury, 2013; Tang et al., 2022). Matrix volume maximization techniques like MaxInfo apply Singular Value Decomposition (SVD) to the deep feature-frame matrix and use MaxVol index selection to identify a subset of pivots that maximize the spanned geometric volume in latent space (Brkic et al., 2026). Similarly, Temporal Segment Density Peaks Clustering (TSDPC) computes local feature density (ρ_i) and higher-density distance (δ_i) across segmented video intervals, automatically identifying decision graph density peaks ($\gamma_i = \rho_i \cdot \delta_i$) as natural keyframes without requiring a hardcoded cluster budget K_{cluster} (Tang et al., 2022).

Graph-based representations model temporal relationships by constructing spatial proximity topologies in feature space (Kuanar, Panda and Chowdhury, 2013). Dynamic Delaunay Graph Clustering constructs a Delaunay triangulation using Principal Component Analysis (PCA) reduced feature vectors as vertices. Inter-cluster edges are iteratively pruned to maximize Global Standard Deviation (GSD) reduction under a Deviation Ratio structural constraint ($DR \geq 1.0$), preserving intra-cluster topology while disconnecting distinct visual concepts:

$$DR(p_i) = \frac{LSD(p_i)}{GSD(DT)} \geq 1.0 \quad (2)$$

where $LSD(p_i)$ denotes the local standard deviation of edge lengths incident to vertex p_i , and $GSD(DT)$ represents the global graph standard deviation (Kuanar, Panda and Chowdhury, 2013). Extracting the medoid frame closest to the centroid of each remaining connected graph component yields a summary that maintains high semantic coverage while avoiding redundant keyframe duplication (Kuanar, Panda and Chowdhury, 2013).

2.2.4 Query-Aware and Multimodal Language-Steered Selection

While unsupervised and pixel-level methods construct generic static summaries, robotic applications operating under specific task instructions benefit from language-conditioned frame selection. Query-aware extraction mechanisms align video candidate frames directly with textual prompts, prioritizing visual samples that contain task-relevant semantic entities or actionable context (S. Zhang et al., 2025).

Direct cross-modal alignment frameworks (such as KeyVideoLLM) project visual candidate frames and user text prompts into a shared multimodal space using pre-trained vision-language encoders like CLIP (Liang et al., 2024). By computing cosine similarity scores between the prompt embedding and each candidate frame embedding, Top- K ranking mechanisms isolate high-scoring visual candidates, ensuring that the visual context provided to downstream foundation models remains tightly focused on query-relevant target entities (S. Zhang et al., 2025).

To overcome the spatial over-clustering and temporal bias associated with naive Top- K ranking, non-parametric probabilistic sampling architectures like BOLT convert cross-modal similarity scores into continuous Probability Density Functions (PDF) (Shuming Liu et al., 2025; S. Zhang et al., 2025). By integrating the similarity PDF into a Cumulative Distribution Function (CDF) and applying Inverse Transform Sampling, keyframes are selected uniformly across the cumulative target probability space. This probabilistic formulation ensures keyframe extraction scales dynamically with query-relevant scene transitions, preserving overall temporal continuity while concentrating visual tokens on task-critical events (Shuming Liu et al., 2025).

2.2.5 Comparative Summary of Video Keyframe Extraction Techniques

The video transformation and frame extraction methodologies reviewed in this section represent a progression from lightweight low-level pixel heuristics to language-steered multimodal selection algorithms. As summarized in Table 2, these techniques balance computational complexity, execution latency, temporal coverage, and semantic alignment.

By filtering redundant temporal visual noise prior to foundation model ingestion, these methods optimize token budget utilization and reduce inference latency on resource-constrained edge platforms. Table 2 provides a structured comparative synthesis of all eighteen keyframe extraction algorithms evaluated, categorizing them by category, method/implementation, feature space, keyframe selection criterion, and relative computational overhead.

Table 2: Comparative Summary of Established Video Keyframe Extraction and Selection Methodologies.

Category	Method / Implementation	Feature Space	Keyframe Selection Criterion	Overhead
Structural & Low-Level	Codec I-Frames	Bitstream Metadata	Container GOP encoder keyint intervals	Negligible
	PySceneDetect (Castellano, 2024)	HSV Color Variance	Adaptive statistical rolling threshold cut	Negligible
	Statistical Thresholding (PeakUtils) (Negri, 2020)	Pixel / Histogram	Absolute difference vs. mean + std offset	Negligible
	SSIM Thresholding	Luminance / Structure	Sequential structural dissimilarity drop ($< T$)	Low
	HSV Histogram	2D HSV Histogram	Bhattacharyya distance max divergence	Low
	Laplacian Sharpness	Gradient Variance	Maximize spatial gradient variance	Low
	Curvature Points (CP)	Color / Edge / Wavelet	Inflection midpoints on cumulative curve	Low
	Bitwise XOR	Binary Pixel Matrix	Maximize bitwise matrix XOR divergence	Negligible
Motion & Kinematic	Covariance Eigenvalues	Local Intensity Matrix	Minimal covariance matrix eigenvalues	Low
	AdaFrame	Sparse Feature Acceleration	Adaptive state memory vector maxima	Medium
Deep Latent & Graph	MGSampler	Dense Optical Flow	Global kinetic optical flow energy peaks	High
	Katna Framework (Lab, 2020)	LUV Color Space	Multi-stage K -Means + blur filter	Medium
	Latent K -Means	CLIP Image Latents	Distance to latent cluster centroid medoids	Medium
	MaxInfo	Deep SVD Subspace	SVD singular value MaxVol pivot selection	High
	TSDPC	Deep Latent Densities	Density peak scores ($\gamma = \rho \cdot \delta$)	Medium
	Delaunay Graph	PCA Feature Graph	Iterative edge pruning under $DR \geq 1.0$	High
Query-Aware VLMs	KeyVideoLLM	CLIP Vision-Language	Top- K cross-modal cosine similarity	Medium
	BOLT	Multimodal CDF Space	Inverse transform sampling on similarity PDF	Medium

2.3 Performance Evaluation Metrics

To rigorously assess the performance of the proposed robotic perception pipeline, a multi-dimensional evaluation framework is employed. This framework evaluates the output of the VLM across three critical dimensions: syntactic structure, semantic fidelity, and visual-spatial grounding.

2.3.1 Syntactic Evaluation: BLEU

The Bilingual Evaluation Understudy (BLEU) serves as the baseline for measuring the syntactic overlap between the agent’s generated description and human-authored reference instructions (Papineni et al., 2002). BLEU calculates a modified n -gram precision p_n to prevent the “overgeneration” of common words. The metric is formulated as the geometric mean of modified n -gram precisions, multiplied by an exponential brevity penalty (BP) to discourage unnaturally short outputs:

$$\text{BLEU} = \text{BP} \cdot \exp \left(\sum_{n=1}^{N_{\text{gram}}} w_n \log p_n \right) \quad (3)$$

where w_n represents uniform weights $1/N_{\text{gram}}$, and p_n is the modified precision for n -grams up to $N_{\text{gram}} = 4$. The brevity penalty is defined as:

$$\text{BP} = \begin{cases} 1 & \text{if } c > r_{\text{ref}} \\ \exp \left(1 - \frac{r_{\text{ref}}}{c} \right) & \text{if } c \leq r_{\text{ref}} \end{cases} \quad (4)$$

where c is the candidate length and r_{ref} is the reference length.

2.3.2 Semantic Fidelity: BERTScore

While BLEU captures surface-level similarity, it fails to account for meaning-preserving lexical diversity such as the use of synonyms which is vital for human-robot interaction. To address this, we utilize BERTScore (T. Zhang et al., 2020), which leverages contextual embeddings from a pre-trained RoBERTa model to compute token-level similarity. For a reference x and candidate $\hat{x}$, BERTScore computes the greedy matching of their token embeddings using cosine similarity:

$$R_{\text{BERT}} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} x_i^\top \hat{x}_j \quad (5)$$

By focusing on the F1 measure derived from these precision and recall scores, the system ensures that robotic commands are evaluated based on their underlying semantic intent rather than exact word-matching.

2.3.3 Visual-Spatial Grounding: CLIPScore

A foundational requirement for robotic perception in complex environments is visual grounding ensuring that a vision-language model’s spatial claims are physically and semantically supported by the input imagery. Traditional metric suites such as METEOR (Banerjee and Lavie, 2005) and SPICE (Anderson, Fernando et al., 2016) rely heavily on human-annotated ground-truth captions. However, in dynamic robotic operating contexts, reference captions are often impractical to collect and fail to capture the real-time geometric reality of the visual scene.

To address reference dependence, modern vision-language literature widely adopts CLIPScore (Hessel et al., 2022), a reference-free visual grounding metric. Rather than comparing candidate text against human references, CLIPScore directly evaluates the alignment between the candidate caption c and the input image I by projecting both modalities into a shared D -dimensional continuous embedding space.

Formally, the visual feature vector $\mathbf{z}_v = E_V(I)$ and textual feature vector $\mathbf{z}_t = E_T(c)$ are extracted via CLIP’s pre-trained vision and text encoders, respectively. Both embeddings are mapped to unit length via L_2 -normalization:

$$\hat{\mathbf{z}}_v = \frac{E_V(I)}{\|E_V(I)\|_2}, \quad \hat{\mathbf{z}}_t = \frac{E_T(c)}{\|E_T(c)\|_2} \quad (6)$$

The semantic visual-textual alignment is quantified using the cosine similarity of the normalized vectors. To prevent negative correlations from distorting score distributions, the cosine similarity is clipped at zero and scaled by a factor w :

$$\text{CLIP-S}(c, I) = w \cdot \max(\hat{\mathbf{z}}_t \cdot \hat{\mathbf{z}}_v, 0) \quad (7)$$

Following the formulation established by Hessel et al. (2022), a rescaling factor of $w = 2.5$ is applied to “stretch” the score distribution across candidate captions, enhancing its discriminative sensitivity to subtle semantic variations. Mechanistically, this formulation functions as an objective visual hallucination check: because ungrounded objects, fabricated spatial relationships, or hallucinated scene attributes shift the direction of $\hat{\mathbf{z}}_t$ away from $\hat{\mathbf{z}}_v$, the dot product decreases proportionately, directly penalizing ungrounded model outputs.

2.3.4 Discussion on Alternative Metrics

While the current framework prioritizes syntactic, semantic, and grounding-based evaluation, several other established metrics were considered, such as METEOR (Banerjee and Lavie, 2005), CIDEr (Vedantam, Zitnick and Parikh, 2015), and SPICE (Anderson, Fernando et al., 2016).

METEOR improves upon BLEU by incorporating stemming and synonym matching; however, its reliance on fixed external paraphrasing dictionaries makes it less flexible than the contextual embedding approach of BERTScore (T. Zhang et al., 2020). **CIDEr** was developed specifically for image captioning to emphasize human-like consensus through TF-IDF weighting, but it requires a large corpus of reference captions to be statistically effective, which is often unavailable in real-time robotic interaction scenarios (Hessel et al., 2022). Furthermore, while **SPICE** excels at evaluating scene graphs and propositional content, its high sensitivity to the output of dependency parsers can introduce noise when processing the concise, technical instructions typical of autonomous navigation. By excluding these in favor of a streamlined Tri-Metric suite, the proposed evaluation remains computationally efficient while maintaining a high correlation with both human judgment and the physical reality of the environment.

3 Methodology

After analyzing current evaluation paradigms and identifying critical vulnerabilities in how vision-language foundation models (VLMs) handle systematic visual perturbations and temporal redundancy, we propose a comprehensive framework comprising tailored image-augmentation and video-keyframe-extraction techniques. A core hypothesis of this work is that no single augmentation or extraction technique is universally optimal; rather, its efficacy is intrinsically linked to the semantic complexity of the text prompt and the structural topology of the visual content. To systematically test this hypothesis and evaluate the methods across diverse levels of visual and textual granularity, we curated a specialized evaluation dataset designed to reveal the performance trade-offs of each approach.

3.1 Task-Aware Image Augmentation Strategies

Visual prompt augmentation transforms raw RGB images into spatially enriched representations, guiding the cross-attention mechanisms of vision-language models toward task-relevant regions (Shtedritski, Christianos and Vedaldi, 2023). To address distinct perceptual bottlenecks, including legibility failures, spatial ambiguity, overlooked safety hazards, and the lack of metric depth, we design four specialized, modular visual-prompt-augmentation techniques. Each method automatically parses the visual scene or query, renders standardized high-contrast spatial prompts, and synthesizes a dual-panel vertical collage ($I_{\text{aug}} \in \mathbb{R}^{H \times W \times 3}$) that pairs the uncompressed original frame with an augmented visual canvas.

3.1.1 Context-Aware Zoom (CAZ)

When humans encounter illegible text, small digital displays, or obscure visual details, the intuitive response is to move closer or zoom in. During VLM inference, fine-grained visual elements are frequently lost when full-resolution images are downsampled to fit the model’s standard input-token grid (Shtedritski, Christianos and Vedaldi, 2023), particularly when the object of interest is far from the camera. *Context-Aware Zoom* (CAZ) resolves this legibility bottleneck by performing automated visual reframing.

Given an input image I and a natural-language user query q , CAZ first employs a lightweight LLM planner f_{planner} (powered by LLaVA-Phi3) to parse the prompt and identify a concrete visual target string s :

$$s = f_{\text{planner}}(q) \in \text{String} \quad (8)$$

Using CLIPSeg zero-shot image segmentation (Lüddecke and Ecker, 2022), the target string s is matched against image I to compute a spatial activation heatmap $A = \sigma(\text{CLIPSeg}(I, s))$. A bounding box enclosing pixel activations above the threshold $\tau_{\text{zoom}} = 200$ is extracted with spatial padding $p = 60$ px:

$$x_1 = \max(0, \min(x_{A>\tau}) - p), \quad x_2 = \min(W, \max(x_{A>\tau}) + p) \quad (9)$$

$$y_1 = \max(0, \min(y_{A>\tau}) - p), \quad y_2 = \min(H, \max(y_{A>\tau}) + p) \quad (10)$$

The target region is cropped to form $I_{\text{crop}} = I[y_1 : y_2, x_1 : x_2]$. Although cropping increases the effective spatial resolution of the target-object tokens, naive zooming discards global contextual information, such as the target’s relative position within the broader environment or its proximity to surrounding obstacles (Shtedritski, Christianos and Vedaldi, 2023). To mitigate this context-loss trade-off, CAZ constructs a vertically stacked dual-panel collage that combines the resized original frame with the zoomed region-of-interest (ROI) crop:

$$I_{\text{CAZ}} = \begin{bmatrix} \text{Resize}(I, W, H/2) \\ \text{Resize}(I_{\text{crop}}, W, H/2) \end{bmatrix} \quad (11)$$

By explicitly isolating the target region and increasing its spatial scale, this mechanism allocates a higher density of visual tokens to fine-grained details, such as small text or distant objects, during the model’s cross-attention computation.

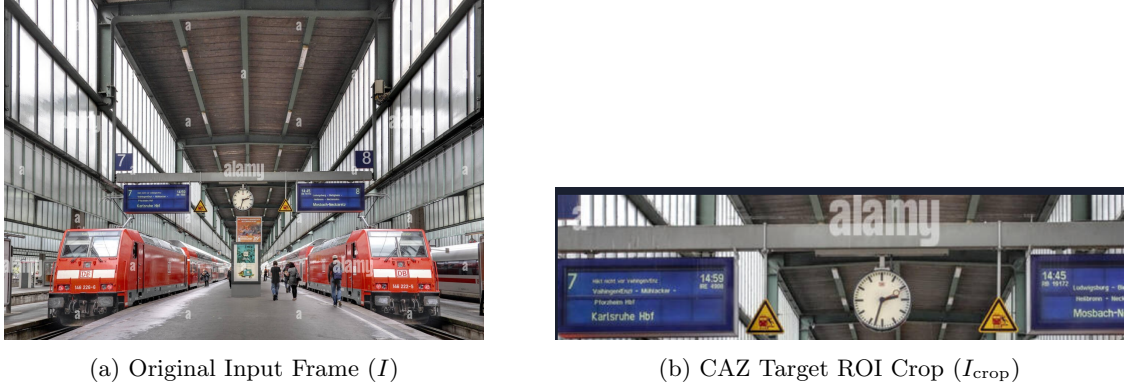


Figure 1: **Context-Aware Zoom (CAZ)** operating on a dense transport hub scene. (a) Original uncropped input frame (I) showing a cluttered train-station platform. (b) Targeted ROI crop (I_{crop}) isolating the overhead schedule board based on query relevance, enhancing visual token density and legibility for fine-grained VLM extraction.

As illustrated in Figure 1, the uncropped image of a cluttered train-station platform (Figure 1a) renders the overhead electronic schedule boards at low pixel density. CAZ processes the fine-grained query about upcoming departures, extracts the target entity (“`schedule board`”), identifies its activation bounds via CLIPSeg, and crops the overhead display region (Figure 1b). This targeted enlargement improves the legibility of crucial textual details, such as the departure time (14:59) and destination (Karlsruhe Hbf), while retaining the spatial context in the original panel.

3.1.2 Query-Aware Bounding Box (QABB)

Although CAZ enhances target resolution, cropping and scaling alter spatial geometry and eliminate the surrounding context from the cropped view. In tasks for which the surrounding environment remains crucial, such as verifying whether a specific screen is attached to a particular control desk, zooming can isolate the target too aggressively.

To address this limitation without altering image scale or discarding spatial context, *Query-Aware Bounding Box* (QABB) preserves global background context by projecting high-contrast bounding overlays directly onto target entities rather than cropping the visual field. Following semantic parsing, $s = f_{\text{planner}}(q)$, and CLIPSeg activation mapping (Lüddecke and Ecker, 2022), regions exceeding the gating threshold $\tau_{\text{qabb}} = 110$ define the target bounds:

$$\Omega_{\text{ROI}} = \{(x, y) \mid A(x, y) > \tau_{\text{qabb}}\} \quad (12)$$

A bright green rectangular overlay ($C_{\text{green}} = [0, 255, 0]$ in BGR format, with thickness $t = 3$ px) is rendered on the image canvas over Ω_{ROI} . The annotated image is then vertically paired with the

unannotated original to form I_{QABB} . By explicitly outlining target entities within their complete environment, QABB directs the VLM’s cross-attention mechanisms toward query-relevant targets while preserving spatial relationships (Shtedritski, Christianos and Vedaldi, 2023).



Figure 2: Qualitative demonstration of **Query-Aware Bounding Box (QABB)**. Given the user query “*What type of plant is shown in this image?*”, the LLM planner identifies the plant as the target entity. QABB projects a high-contrast green bounding box around the activated region, guiding VLM attention while preserving the global context of the occluding robot and the kitchen environment.

As demonstrated in Figure 2, a query may require reasoning about a specific background object, such as classifying the plant behind the embodied agent. Applying a hard crop in this case would remove important spatial and depth cues, including the robot’s partial occlusion of the foliage. The QABB pipeline instead localizes the target entity and overlays a high-contrast, non-destructive bounding box (Figure 2b). This visual prompt anchors the VLM’s cross-attention mechanisms to the plant without altering the image scale or discarding the surrounding geometry of the kitchen environment.

3.1.3 Semantic Hazard Isolation (SHI)

Whereas CAZ and QABB focus on target legibility and object localization, autonomous robotic manipulation and navigation systems remain vulnerable to visual safety anomalies that fall outside conventional closed-vocabulary object classes (Zhiwei Wu et al., 2024; Y. Zhou et al., 2025). An embodied agent or mobile robot operating in an unconstrained environment might fail to perceive a subtle wet-floor spill, sharp glass shards, or irregular physical hazards because standard bounding-box detectors are restricted to predefined dataset categories (Zhiwei Wu et al., 2024). VLMs may also encounter perceptual bottlenecks when small spatial hazards blend into background textures or lack visual contrast (Y. Zhou et al., 2025).

Semantic Hazard Isolation (SHI) addresses this safety vulnerability by converting zero-shot, open-vocabulary hazard segmentations into high-contrast visual prompts (Lüddecke and Ecker, 2022). Rather than relying on closed-vocabulary object classifiers, SHI uses CLIPSeg to evaluate input frames against prompt-specific queries or open-ended safety descriptors ($q_{\text{hazard}} = \text{“something dangerous, safety hazard, sharp object, spill”}$). To remove spurious pixel noise and smooth spatial boundaries, the resulting dense activation heatmap A is binarized at the threshold $\tau_{\text{hazard}} = 95$ and refined through morphological closing with a rectangular structuring element $K_{11 \times 11}$:

$$M_{\text{binary}} = (A > \tau_{\text{hazard}}) \oplus K_{11 \times 11} \ominus K_{11 \times 11} \quad (13)$$

For identified hazard regions where $M_{\text{binary}}(x, y) = 1$, a semitransparent red warning layer ($C_{\text{red}} = [0, 0, 240]$ in BGR format, with $\alpha = 0.6$ and $\beta = 0.4$) is blended directly onto the visual canvas:

$$I_{\text{hazard}}(x, y) = \alpha \cdot I(x, y) + \beta \cdot C_{\text{red}} \quad (14)$$

In addition to the red overlay, major hazard contours with areas exceeding 200 px^2 are enclosed by bold red bounding boxes ($t = 4 \text{ px}$). Stacking the resulting annotated canvas beneath the

original frame yields I_{SHI} . By explicitly highlighting open-vocabulary safety anomalies, SHI encourages downstream VLMs to prioritize physical safety constraints and environmental hazards during spatial reasoning and task planning (Y. Zhou et al., 2025).



(a) Original Input Frame (I)



(b) SHI Annotated Frame (I_{SHI})

Figure 3: **Semantic Hazard Isolation (SHI)** highlighting an unstructured pile of shattered glass and metal debris. Zero-shot open-vocabulary segmentation renders a semitransparent red mask and perimeter border over the hazard zone to guide physical obstacle avoidance.

As illustrated in Figure 3, unconstrained outdoor environments may contain severe physical hazards, such as mounds of broken glass, shattered window frames, and sharp structural debris blocking entrance pathways (Figure 3a). Standard closed-vocabulary object detectors typically lack pre-defined categories for non-rigid piles of mixed debris. SHI instead uses zero-shot, open-vocabulary segmentation to localize the primary hazard footprint (Figure 3b). The semitransparent red warning mask and perimeter bounding box highlight the hazardous surface area, providing downstream VLMs with high-contrast visual cues for obstacle avoidance and safe path planning.

3.1.4 Perception Metric Grounding (PMG)

While the preceding strategies improve semantic localization and visual legibility, standard VLMs lack explicit 3D depth information (Bhat et al., 2023). A model may infer relative visual relationships (e.g., “the chair is behind the desk”) but cannot reliably answer quantitative spatial queries (e.g., “Is the obstacle within 1.2 m of the robot gripper?”).

Perception Metric Grounding (PMG) bridges the gap between 2D pixel semantics and 3D physical geometry by embedding estimated metric distances directly into the visual input stream. PMG integrates a two-stage computer-vision pipeline:

1. **2D Bounding-Box Localization:** YOLOv8 (Jocher, Chaurasia and Jing, 2023) detects discrete COCO object instances and generates bounding coordinates $B = \{b_1, b_2, \dots, b_N\}$.
2. **Metric Depth-Map Estimation:** ZoeDepth (Bhat et al., 2023) predicts a dense monocular metric depth map $D_{\text{metric}} \in \mathbb{R}^{H \times W}$, in which each pixel value represents the estimated distance from the camera in meters.

To prevent boundary noise or background leakage from skewing the distance estimates, PMG computes the median depth within each localized bounding region:

$$d_i = \text{Median}(\{D_{\text{metric}}(x, y) \mid (x, y) \in b_i\}) \quad (15)$$

A green heads-up-display (HUD) overlay, comprising a bounding border and an explicit metric-distance label (e.g., “1.8 m”), is drawn around each detected object. Stacking the resulting

HUD-annotated image beneath the unannotated original yields I_{PMG} , supplying the downstream VLM with explicit quantitative grounding.



(a) Original Input Frame (I)



(b) PMG HUD Grounded Frame (I_{PMG})

Figure 4: **Perception Metric Grounding (PMG)** overlaying HUD depth labels across workstation entities. Fusing YOLOv8 detection with ZoeDepth monocular estimation renders quantitative distance tags (e.g., plant at 1.4 m, laptop at 2.5 m, seating at 7.8 m) directly within the visual stream.

As illustrated in Figure 4, embodied tasks in structured workspaces require fine-grained metric information to distinguish immediate action candidates from distant background clutter. Although standard 2D input frames (Figure 4a) provide rich visual semantics, downstream VLMs may struggle to infer absolute scale or depth ordering among repeated objects, such as identical office chairs along a row of desks. The PMG pipeline computes a dense monocular depth map and overlays localized median-distance labels on detected entities (Figure 4b). This HUD-style visual prompt explicitly grounds foreground elements (1.4 m), interactive objects (2.5 m), and distant targets (7.8 m), providing the VLM with quantitative spatial information for metric reasoning and motion planning.

3.2 Prompt-Guided Keyframe Selection and Visual Grounding

Extending visual prompt augmentation from static two-dimensional images to continuous video streams introduces severe temporal redundancy, storage overhead, and processing bottlenecks. To isolate informative visual states from raw video, traditional video processing literature relies on four core extraction paradigms: pixel-level structural dissimilarity (e.g., SSIM and Laplacian variance (Z. Wang et al., 2004)), motion and kinematic field tracking (such as Lucas-Kanade optical flow (Horn, 1986)), unsupervised deep latent clustering (e.g., K -Means and TSDPC density peaks), and multimodal query-conditioned samplers.

While these conventional approaches demonstrate high efficacy in fixed-camera surveillance and video summarization, empirical benchmarking reveals that they systematically fail on mobile robotics platforms. Structural pixel heuristics are hypersensitive to ambient lighting shifts, flooding extraction pipelines with near-identical frames. Kinematic optical flow trackers suffer from camera ego-motion: as a mobile platform rotates or moves, global background displacement is misinterpreted as meaningful local motion spikes. Furthermore, deep latent clustering enforces rigid frame allocations that often extract dense temporal “bursts”: saving dozens of consecutive, redundant frames of the same target captured fractions of a second apart. This deluge of redundant images rapidly saturates processing buffers, consumes excessive memory on edge hardware, and obscures critical state transitions.

To resolve these challenges, we establish a dual-paradigm keyframe extraction framework that combines hardware-accelerated stream demuxing with query-conditioned visual grounding, structured modularly as illustrated in Figure 5.

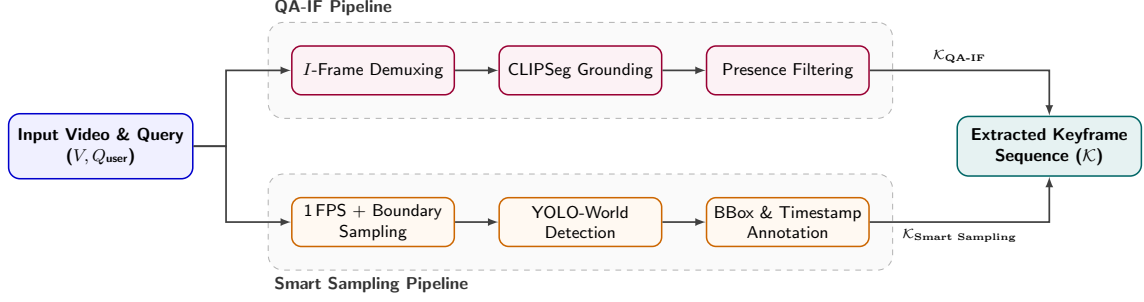


Figure 5: Overview of the Query-Aware Keyframe Selection framework detailing the parallel QA-IF and Smart Sampling extraction pipelines.

3.2.1 Codec-Level *I*-Frame Demuxing with Dense Grounding (QA-IF)

To establish a computationally lightweight foundation, the first extraction paradigm bypasses raw pixel-difference heuristics and operates directly on the underlying video compression container. Modern video codecs (such as H.264/AVC and H.265/HEVC) compress temporal sequences into macroblock hierarchies comprising Intra-coded frames (*I*-frames), Predicted frames (*P*-frames), and Bi-directionally predicted frames (*B*-frames). Formally, an *I*-frame $I_{\text{frame}} \in \mathbb{R}^{H \times W \times 3}$ is encoded as a spatially standalone image using intra-frame spatial Discrete Cosine Transform (DCT) quantization, completely independent of prior or subsequent temporal context:

$$I_{\text{frame}}(x, y) = \mathcal{Q}^{-1}(\text{IDCT}(\mathbf{C}_{\text{intra}}(x, y))) \quad (16)$$

where $\mathbf{C}_{\text{intra}}$ represents the quantized intra-spatial transform coefficients, IDCT denotes the Inverse Discrete Cosine Transform, and $\mathcal{Q}^{-1}$ is the inverse quantization operator. Conversely, inter-coded *P*-frames and *B*-frames encode directional motion vectors $\mathbf{v}_{\text{motion}}$ and spatial residual error deltas $\mathbf{R}(x, y)$ relative to reference frames:

$$P_{\text{frame}}(x, y) = I_{\text{frame}}(x + v_x, y + v_y) + \mathbf{R}(x, y) \quad (17)$$

Decoding intermediate *P*- and *B*-frames requires computationally expensive temporal reconstruction loops and frequently introduces motion blur artifacts into visual processing pipelines. In contrast, demuxing raw *I*-frames directly at the container level yields instantaneous $\mathcal{O}(1)$ packet parsing, providing a candidate frame pool $\mathcal{F}_I = \{I_1, I_2, \dots, I_M\}$ characterized by maximum spatial sharpness and zero temporal motion-blur propagation.

To make these structural *I*-frames query-aware, we route the user prompt Q_{user} through an LLM entity parser f_{planner} (e.g., **LLaVA-Phi3** / **Phi-3 Mini**) to isolate the primary visual target noun $s = f_{\text{planner}}(Q_{\text{user}})$, stripping out extraneous verbs and contextual modifiers. Each candidate frame $I_i \in \mathcal{F}_I$ is paired with s and processed through a CLIPSeg dense spatial grounding backbone (Lüddecke and Ecker, 2022). The model computes an open-vocabulary spatial probability map:

$$P_i(x, y) = \sigma(\text{CLIPSeg}(I_i, s)(x, y)) \in [0.0, 1.0] \quad (18)$$

where $\sigma(\cdot)$ denotes the standard element-wise sigmoid activation function. To suppress localized noise artifacts and prevent single-pixel outliers from corrupting frame ranking, we define the presence score $S(I_i, s)$ as the mean activation of the top- K highest-probability pixels $\Omega_K \subset \mathbb{R}^{H \times W}$:

$$S(I_i, s) = \frac{1}{K} \sum_{(x, y) \in \Omega_K} P_i(x, y) \quad (19)$$

Candidate frames are strictly filtered against a presence threshold $\tau_{\text{presence}} \in [0.0, 1.0]$:

$$\mathcal{F}_{\text{match}} = \{I_i \in \mathcal{F}_I \mid S(I_i, s) \geq \tau_{\text{presence}}\} \quad (20)$$

If no frames satisfy τ_{presence} , the frame candidate set is dropped to avoid propagating false-positive scene context. Otherwise, surviving frames are ranked by descending visual presence score $S(I_i, s)$, truncated to the top N_{max} candidates, and restored to their original chronological sequence order to preserve temporal consistency for downstream multimodal reasoning.



Figure 6: Visual demonstration of the Query-Aware *I*-Frame extraction (QA-IF) pipeline. Given the user query “How many flower pots do you see in the room?”, the candidate container *I*-frames are extracted at codec level (a). The semantic grounding module isolates the specific *I*-frame containing the plant shelf and target potted vegetation (b) while discarding uninformative perspectives.

Figure 6 illustrates this query-conditioned extraction process. Given the input query “How many flower pots do you see in the room?”, the planner isolates the target entity “flower pots”. The candidate container *I*-frame pool (Figure 6a) is densely parsed, and the spatial activation backbone isolates the specific *I*-frame capturing the plant shelf and potted vegetation (Figure 6b), effectively eliminating uninformative room angles and preventing downstream visual token bloat.

3.2.2 Open-Vocabulary Object Grounding and Visual Prompting (Smart Sampling)

While dense pixel-level segmentation via CLIPSeg provides precise heatmap coverage, it can be computationally heavy during real-time extraction (< 10 frames per second (FPS)) and produces soft activation masks that lack explicit bounding coordinates. To achieve real-time keyframe filtering (> 50 FPS) with explicit spatial-temporal visual cues, the second extraction paradigm couples uniform/boundary temporal sampling with open-vocabulary object detection and visual prompt overlays.

Given an input video sequence of duration T , candidate frames are sampled at a fixed temporal frequency of 1Hz augmented with boundary state frames (t_0 and t_{end}), yielding candidate set $\mathcal{F}_{\text{sample}}$. Concurrently, the user prompt Q_{user} is parsed via lightweight natural language chunking to extract target entity classes $S = \{s_1, s_2, \dots, s_m\}$. To eliminate per-frame text-encoding latency, candidate frames are processed through **YOLO-World** (Cheng et al., 2024), an open-vocabulary detector that evaluates target text embeddings against visual feature maps in real time:

$$\mathbf{B}_i = \text{YOLO-World}(I_i, S, \tau_{\text{conf}}) \quad (21)$$

where $\mathbf{B}_i = \{(x_1, y_1, x_2, y_2, c, \text{conf})\}$ represents the set of detected bounding boxes, class labels $c \in S$, and detection confidence scores $\text{conf} \geq \tau_{\text{conf}}$ (with $\tau_{\text{conf}} = 0.25$).

To maximize visual feature clarity and spatial interpretability, surviving frames undergo **Visual Prompt Overlay** inspired by the Set-of-Mark paradigm. For each selected frame I_i , bounding boxes from $\mathbf{B}_i$ are rendered directly onto the visual canvas alongside a temporal timestamp badge indicating the exact frame timestamp t_i . To enforce strict frame-budget constraints ($K_{\text{QA}} \leq 3$), frames are sorted by maximum detection confidence and temporal NMS is applied to prune redundant consecutive detections of stationary objects.

3.2.3 Keyframe Output and Pipeline Synthesis

The surviving query-aware keyframes from either pipeline ($\mathcal{K} = \{k_1, k_2, \dots, k_K\}$) represent the minimal, highly informative visual subset required to fulfill the extraction criteria. By isolating key temporal events and rendering explicit bounding boxes and timestamp metadata directly onto the image canvas, the extraction pipeline drastically reduces video redundancy while preserving essential spatial and temporal details for downstream processing.

3.3 Evaluation Strategies and Computational Infrastructure

Traditional lexical metrics such as BLEU and ROUGE evaluate n -gram overlap, while BERTScore compares contextual embeddings; however, these automated metrics frequently fail to capture functional equivalence, spatial reasoning, and compositional logic in open-ended Vision-Language Model (VLM) predictions. To overcome these limitations, the framework implements a two-pronged evaluation strategy: real-time latency and token profiling executed directly on the edge platform (NVIDIA Jetson AGX Thor), paired with a deep semantic evaluation pipeline executed offline on a High-Performance Computing (HPC) workstation.

Offloading semantic evaluation to an HPC environment equipped with dual NVIDIA RTX 4090 GPUs (48 GB aggregate VRAM) allows us to deploy high-capacity reasoning models and run multi-agent debate consensus loops across thousands of dataset rows concurrently. This dual-hardware approach ensures that complex Chain-of-Thought (CoT) reasoning models do not starve the edge platform of memory or computational resources during live inference benchmarking. Furthermore, utilizing local open-weight model instances such as DeepSeek-R1 32B (DeepSeek-AI et al., 2025), Qwen2.5 14B (Qwen Team, 2024), and Llama 3 (Dubey et al., 2024) via Ollama rather than cloud-based APIs (e.g., GPT-4) prevents evaluation drift caused by unannounced cloud model updates, guaranteeing stable and reproducible evaluation baselines (Zheng et al., 2023).

3.3.1 LLM-as-a-Judge Semantic Correctness Methodology

To evaluate open-ended textual predictions without relying solely on surface-level word matching, we deploy an automated *LLM-as-a-Judge* semantic evaluation pipeline (Zheng et al., 2023). We select a reasoning-focused model **DeepSeek-R1 (32B)** (DeepSeek-AI et al., 2025) as the primary evaluator due to its strong instruction compliance and explicit CoT capabilities. Visual-spatial reasoning tasks require step-by-step logic; DeepSeek-R1 generates internal thinking paths that verify spatial geometry, object presence, and physical context before assigning a final correctness score.

To provide granular scoring, predictions y_{pred} are assessed against ground-truth references y_{gt} while explicitly conditioning on the original user query q . The judge applies a ten-tier grading scale ranging from $[0, 9]$ (completely incorrect or hallucinatory) to $[90, 100]$ (semantically equivalent and factually aligned).

Because DeepSeek-R1 generates explicit internal reasoning steps enclosed within `<think>...</think>` tags, the parser strips these reasoning blocks using regular expressions prior to score extraction:

$$y_{\text{cleaned}} = \text{re.sub}(\text{"i?think_.*?i?/think_"}, \text{"", } y_{\text{raw}}) \quad (22)$$

To reduce generation noise and minor output variation in autoregressive language models, we adopt an ensembling approach inspired by multi-judge evaluation architectures (Verga et al., 2024). The semantic evaluation function is executed for $N_{\text{pass}} = 3$ parallel passes per sample. The final semantic-correctness score S_{eval} is the arithmetic mean across these passes:

$$S_{\text{eval}} = \frac{1}{N_{\text{pass}}} \sum_{i=1}^{N_{\text{pass}}} f_{\text{LLM}}(q, y_{\text{gt}}, y_{\text{pred}}; T_{\text{sample}} = 0.2) \quad (23)$$

where f_{LLM} represents the structured evaluation prompt executed with low LLM sampling temperature ($T_{\text{sample}} = 0.2$).

3.3.2 Heterogeneous Multi-Agent Debate Methodology

When evaluating 35 competing visual prompt augmentation methods across identical images, relying on a single evaluator prompt can introduce position, length, or verbosity bias. To overcome these single-evaluator failure modes, we implement a **Heterogeneous Multi-Agent Debate Framework** executed on the dual RTX 4090 HPC workstation. Figure 7 illustrates the multi-stage agentic workflow and consensus voting pipeline.

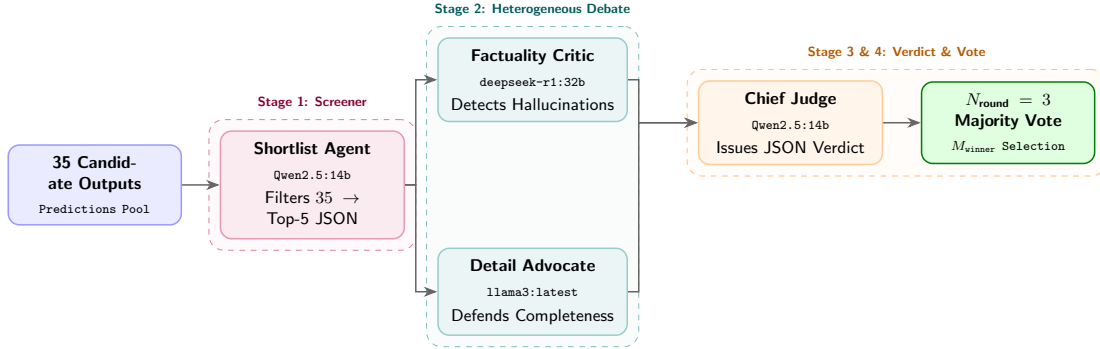


Figure 7: Heterogeneous multi-agent debate architecture illustrating candidate shortlisting, parallel agent debate (Critic vs. Advocate), Chief Judge verdict generation, and $N_{\text{round}} = 3$ majority consensus voting.

This framework distributes evaluation responsibilities across specialized LLM agent roles based on individual model architectural strengths:

1. **Candidate Shortlisting Agent (Qwen2.5:14b):** Comparing 35 full text responses simultaneously exceeds the context window or attention focus of most evaluators. A fast 14B parameter model acts as a preliminary screener, filtering the candidate pool down to the Top-5 methods ($K_{\text{shortlist}} = 5$) matching the ground truth and returning a structured JSON list.
2. **Factuality Critic Agent (deepseek-r1:32b):** The shortlisted candidates are passed to a reasoning-focused model acting as a strict critic. The Critic inspects candidate answers specifically for visual hallucinations, false statements, or physical contradictions relative to the ground truth.
3. **Visual Detail Advocate Agent (llama3:latest):** Concurrently, a separate model evaluates the same candidates for spatial completeness and thoroughness, defending the most detailed prediction that remains factually accurate.
4. **Chief Judge Agent (Qwen2.5:14b):** A neutral Judge model reviews the original query, ground truth reference, candidate predictions, and written arguments from both the Critic and Advocate. The Judge emits a JSON verdict containing the winning method string and concise explanatory reasoning.

To guarantee decision stability across stochastic generations, the entire multi-agent debate workflow is repeated across $N_{\text{round}} = 3$ independent passes for every image in the dataset. A majority consensus vote determines the final winning augmentation method:

$$M_{\text{winner}} = \text{Mode}(\{\text{DebateRun}_1, \text{DebateRun}_2, \text{DebateRun}_3\}) \quad (24)$$

By isolating this heavy multi-agent debate loop within the HPC infrastructure, we obtain unbiased method rankings across large-scale dataset runs without slowing down real-time edge processing on the robot.

3.4 Dataset Collection and Quantitative Benchmark Analysis

To systematically evaluate VLMs across the cognitive hierarchy required for mobile robotics, we established a benchmark framework spanning seven functional query categories: **Navigation & Path Planning**, **Object Detection & Localization**, **Spatial Reasoning & Layout Analysis**, **Hazard & Obstacle Identification**, **Human Presence & Interaction (HRI/HOI)**, **Contextual & Scene Understanding**, and **Multi-Hop Compositional Reasoning**. This taxonomy reflects the operational stack of mobile autonomous systems, ranging from low-level spatial geometry and collision avoidance (Sünderhauf et al., 2018), through mid-level path planning and spatial grounding (Anderson, Q. Wu et al., 2018), to high-level semantic scene understanding and sequential logic (Shah et al., 2022).

3.4.1 Static Image Benchmark Corpus

To construct a static visual benchmark tailored to these seven categories, we sourced images from diverse open-access repositories, including *SuperGlasses* (Jiang et al., 2026), *GuideDog* (Kim et al., 2026), *Urban1k* (B. Zhang et al., 2024), *NaviTrace* (Windecker et al., 2025), *PhotoChat* (Zang et al., 2021), *ADE20K* (B. Zhou et al., 2018), and the *LOw-Light (LOL)* dataset (Wei et al., 2018). Rather than downloading complete repository dumps (which range from 70–150 GB per dataset), targeted subsampling was employed to retrieve representative batches from each source. This strategy maximized situational diversity while remaining within local storage and computational constraints. Including degraded-illumination environments from the *LOL* dataset enabled evaluating VLM robustness under harsh shadows and nighttime conditions where standard visual pipelines degrade (Wei et al., 2018). In total, this sampling strategy yielded a consolidated repository of **731 unique images**: 661 well-lit operational scenes and 70 low-light frames (9.58%).

To establish a standardized question-answering benchmark for the image repository, we developed an automated question-generation and consensus-based reference-annotation pipeline powered by *Qwen2.5-VL-32B*. For each image, the model generated evaluation questions drawn from the seven functional domains. To reduce reliance on manual annotation and eliminate single-pass generation bias, a multi-run consensus mechanism was implemented: *Qwen2.5-VL-32B* independently generated three candidate answers (A_1, A_2, A_3) per question across distinct inference passes. Pairwise sentence-level BLEU scores were computed for each candidate triplet, selecting the candidate with the highest mean consensus score as the ground-truth reference answer:

$$A_{\text{GT}} = \arg \max_{A_i \in \{A_1, A_2, A_3\}} \frac{1}{2} \sum_{j \neq i} \text{BLEU}(A_i, A_j) \quad (25)$$

In total, this pipeline produced a benchmark of **2,103 distinct question-answer pairs** across the 731 visual scenes. The evaluation questions have an average length of 17.72 ± 5.64 words, while the consensus reference answers provide detailed visual descriptions averaging 139.74 ± 93.70 words. A total of 128 QA pairs (6.09%) explicitly evaluate low-light operational scenarios.

Table 3: Distribution and BLEU-Based Consensus Metrics Across Question Categories in the Static Image Benchmark Repository.

Question Category	Total QA Pairs	Share (%)	Mean Consensus BLEU	Median BLEU
Contextual & Scene Understanding	683	32.48%	0.4557 ± 0.0989	0.4531
Spatial Reasoning & Layout Analysis	365	17.36%	0.5081 ± 0.1659	0.4731
Object Detection & Localization	329	15.64%	0.6117 ± 0.1759	0.6056
Hazard & Obstacle Identification	265	12.60%	0.5039 ± 0.1129	0.4927
Human Presence & Interaction (HRI/HOI)	199	9.46%	0.5254 ± 0.1722	0.4948
Multi-Hop Compositional Reasoning	196	9.32%	0.5576 ± 0.1744	0.5511
Navigation & Path Planning	66	3.14%	0.5041 ± 0.1329	0.4822
Total Image Benchmark Corpus	2,103	100.0%	0.5129 ± 0.1528	0.4849

As summarized in Table 3, the consensus pipeline achieved an overall mean BLEU score of 0.5129 ± 0.1528 (median 0.4849). The highest consensus agreement occurred in *Object Detection & Localization* (BLEU = 0.6117), reflecting the constrained nature of object bounding descriptions. Conversely, *Contextual & Scene Understanding* exhibited the lowest mean score (0.4557), owing to stylistic variations in open-ended scene summaries.

3.4.2 Dynamic Video Benchmark Corpus

To extend evaluation to dynamic visual streams and assess temporal keyframe selection strategies, we curated a dedicated repository of **70 high-resolution video sequences** sourced from open-access platforms (*Pexels* and *Pixabay*). To maintain structural parity with the static image benchmark, the video repository is strictly balanced across the identical seven functional categories, allocating exactly 10 targeted video clips to each domain. The video sequences range from 4.43 to 63.60 seconds in duration, capturing real-world mobile environments such as warehouse aisles, public transit hubs, busy urban intersections, and residential corridors.

For each video sequence, category-specific evaluation prompts and comprehensive ground-truth reference descriptions were synthesized using *Gemini 3.6 Flash*. To ensure absolute factual precision and eliminate vision-language hallucinations, all generated question-answer pairs underwent rigorous human-in-the-loop verification and manual refinement.

To evaluate keyframe extraction techniques and benchmark downstream inference latency, every video scenario is annotated with temporal keyframe metadata. This includes identifying the minimal informative frame subset ($N_{\min}$) and exact keyframe timestamps (T_{key}) required to satisfy the query (Q_{user}). Across the 70 video evaluation scenarios, static spatial queries (e.g., *Object Detection & Localization*) require a minimal sampling density ($N_{\min} = 1.10 \pm 0.31$ frames), whereas dynamic and compositional queries (e.g., *HRI/HOI* and *Multi-Hop Compositional Reasoning*) require temporal multi-frame tracking ($N_{\min} = 2.10 \pm 0.42$ frames at key transition timestamps). This dual-layer structure enables unified benchmarking of both sparse keyframe sampling algorithms and downstream VLM reasoning performance.

3.5 System Architecture and Software Engineering

To systematically evaluate VLMs under visual perturbations and dynamic video keyframe sequences, we designed a modular, high-throughput software platform. The system decouples interactive client interfaces, asynchronous server execution, computer vision transformation pipelines, and foundation model inference bridges into clean, independent modules. Figure 8 illustrates the high-level system architecture and data pipeline.

3.5.1 Repository Codebase Structure

The repository organizes backend routes, computer vision modules, extensible research hooks, and client-side interfaces into clean, standalone directories:

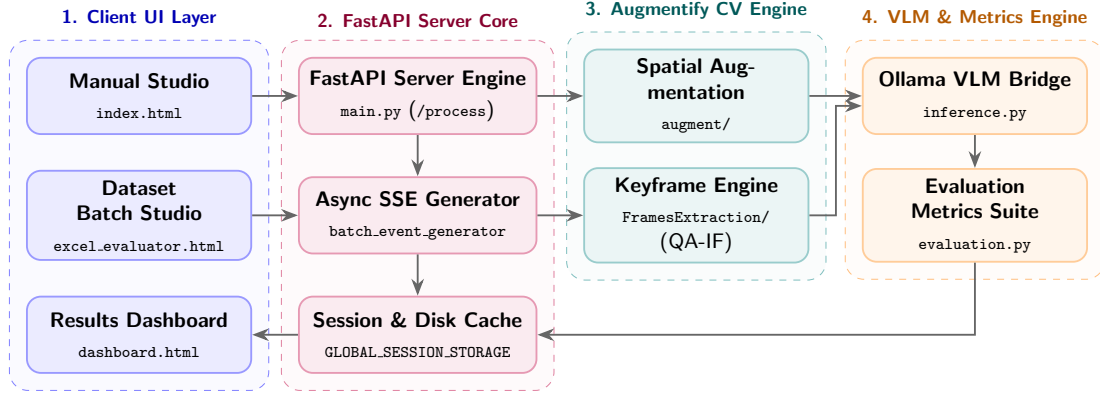


Figure 8: System component diagram showcasing end-to-end data processing from interactive client web studios, through the FastAPI server engine and Augmentify computer vision framework, to local VLM inference and metric evaluation suites.

```

backend/
|-- main.py                # FastAPI core router & SSE event generator
|-- config.py             # Supported model & augmentation registry
|-- inference.py          # Local Ollama VLM inference bridge
|-- evaluation.py         # Comprehensive evaluation metrics suite
|-- Augmentify/          # Standalone Computer Vision Library
|   |-- augment/         # Spatial visual prompt engines (CAZ, QABB, SHI, PMG)
|   |-- FramesExtraction/ # Video keyframe samplers (QA-IF, Smart Sampling, Optical Flow)
|   |-- Futurework/      # Abstract templates & experimental stubs
|   |-- examples/        # Example scripts & algorithm usage demos
|   |-- README.md        # Library specifications & developer documentation
|-- frontend/            # Interactive web client tools
|   |-- index.html       # Single-item evaluation studio
|   |-- excel_evaluator.html # Automated dataset batch runner (.xlsx/.csv)
|   |-- dashboard.html   # Analytics & results export studio
|-- static/              # Cached images, generated logs, & session dumps

```

3.5.2 Asynchronous Streaming Backend Engine (main.py)

The application server is built on FastAPI and uses Server-Sent Events (SSE) streaming (`text/event-stream`) to maintain continuous, non-blocking communication with client applications. Upon receiving batch requests at the `/process` route, the server spawns an asynchronous event generator (`batch_event_generator`). This streams real-time execution progress, status logs, and intermediate frame data back to the browser, mitigating HTTP timeout risks during heavy visual processing or multi-frame inference passes.

For each evaluation item, the engine processes media through two concurrent execution paths:

- Baseline Evaluation Path:** Routes the original, unaugmented visual input directly to the Vision-Language Model. This establishes reference baseline metrics, measuring original model precision (y_{orig}), output word count, and base latency (t_{orig}).
- Augmented Evaluation Path:** Passes the visual input through the Augmentify library to apply targeted spatial transformations or extract keyframes. The resulting augmented asset is processed by the model to record augmented latency (t_{proc}), token outputs, and accuracy deltas (ΔScore).

To safeguard against data loss during long execution runs, session records are maintained in-memory within a global state storage object (`GLOBAL_SESSION_STORAGE_MATRIX`). Intermediate

outputs are asynchronously appended as CSV records under `../static/cache/`, allowing completed runs to be restored seamlessly if the server restarts.

3.5.3 Modular Computer Vision Library (Augmentify)

To ensure computer vision routines remain decoupled from backend web logic, all spatial transformation and keyframe algorithms are structured as an independent Python package named **Augmentify**. The library exposes a standardized functional interface:

$$I_{\text{augmented}} = \text{apply_augmentation}(I_{\text{input}}, \text{method}, Q_{\text{user}}) \quad (26)$$

This design isolates web API endpoints from underlying vision backbones (e.g., CLIPSeg, ZoeDepth, YOLO-World, and FastSAM). The main entry point accepts a PIL Image or video stream alongside a user prompt Q_{user} , executes the designated visual prompt algorithm, and returns the modified image tensor ready for downstream inference.

3.5.4 Extensibility and Future Research Framework (Futurework)

To facilitate ongoing research and ease the integration of emerging computer vision paradigms, the **Augmentify** package incorporates a dedicated **Futurework/** module. This directory serves as a standardized sandbox containing lightweight interface stubs, empty function signatures, and fallback mechanics (such as returning identity image tensors or uniform random video frame selections).

The **Futurework/** module provides abstract contracts and developer documentation outlining future research directions:

- **Dynamic 3D Gaussian Splatting Prompts:** Template methods designed to receive neural radiance fields (NeRFs) or 3D scene point clouds, allowing visual prompt overlays to be rendered in 3D camera space prior to 2D image projection.
- **Reinforcement Learning Prompt Policy Optimization:** Abstract hooks intended for training policy agents that dynamically select optimal augmentation methods (e.g., deciding between *Context-Aware Zoom* and *Semantic Hazard Isolation*) based on input prompt ambiguity.
- **Adaptive Codec-Aware Keyframe Filtering:** Experimental templates for directly parsing modern video containers (AOMedia Video 1 (AV1), Versatile Video Coding (VVC/H.266)) to evaluate macroblock motion vectors without decoding video packets.

3.5.5 Vision-Language Model Bridge (inference.py)

The inference module standardizes communications between the evaluation platform and locally hosted VLMs (e.g., Qwen2.5-VL, LLaVA-Phi3) interfaced via local Ollama instances (`ollama.chat`). `inference.py` validates local asset paths, constructs structured prompt payloads, handles image encoding, and enforces deterministic generation parameters ($T_{\text{gen}} = 0.0$), gracefully returning structured response text and raw generation timestamps back to the core server.

3.5.6 Comprehensive Evaluation Metrics Suite (evaluation.py)

Upon receiving generated model predictions for both unaugmented baseline (y_{orig}) and visual prompt-augmented (y_{proc}) image paths, the automated evaluation pipeline (`evaluation.py`) assesses model outputs against ground-truth reference annotations (y_{gt}) and physical input canvases (I). To execute multi-metric evaluation directly on the NVIDIA Jetson AGX Thor edge platform

without inducing memory fragmentation or reloading overhead, the module implements a lazy-loaded singleton design pattern for heavyweight neural evaluators. The evaluation suite assesses performance across five core analytical dimensions:

1. **Inference Latency Delta (Δt):** Precision timers capture the wall-clock execution duration in milliseconds for baseline (t_{orig}) and augmented (t_{proc}) inference passes. The net computational latency cost introduced by visual prompt transformations is isolated by computing the absolute duration difference between execution runs.
2. **Contextual Embedding Alignment (BERTScore F1):** Deep semantic similarity between generated predictions and ground-truth references is computed via contextual token embeddings. To eliminate PyTorch weight initialization delays across benchmark runs, the module utilizes a cached singleton instance (`BERTScorer`) leveraging a lightweight `distilbert-base-uncased` backbone.
3. **Lexical N-Gram Precision (BLEU Score):** Surface-level word agreement is quantified using sentence-level BLEU with Chen-Cherry Smoothing (Method 1). This smoothed n-gram overlap measure evaluates concise phrase agreement without excessively penalizing minor syntactic variations or phrasing reorderings.
4. **Visual-Spatial Cross-Modal Grounding (CLIPScore):** Reference-free visual grounding is computed to measure cross-modal alignment between the generated text description and the physical visual canvas (I). Leveraging a singleton instance of `openai/clip-vit-base-patch32` initialized on the local computing device, L_2 -normalized image and text feature embeddings are extracted to calculate cosine similarity with a standard scaling factor ($w = 2.5$). This verifies whether visual prompt overlays preserve cross-modal fidelity and prevent text-induced visual hallucinations.
5. **Token Generation Statistics:** The module tracks output generation volume by calculating exact whitespace token counts for baseline (N_{orig}) and augmented (N_{proc}) model outputs, isolating how visual prompt overlays influence VLM output conciseness and autoregressive decoding iterations.

All metric passes are aggregated by a central execution function (`evaluate_outputs`) into structured JSON payloads consumed by downstream multi-agent judge panels and statistical telemetry dashboards.

3.5.7 Client Workflows, Logging, and Data Export

The user interface layer provides ad-hoc evaluation tools, batch benchmarking utilities, and complete data export capabilities:

- **Manual Evaluation Studio (`index.html`):** Enables single-item inspection. Users upload image or video files, select visual augmentation or keyframe strategies, and enter target prompts (Q_{user}) alongside ground-truth text (y_{gt}) to observe visual transformations and response outputs in real time.
- **Dataset Batch Studio (`excel_evaluator.html`):** Enables automated, multi-item evaluation. Utilizing client-side JavaScript (`SheetJS`), it reads uploaded `.xlsx` or `.csv` files, offering interactive column-mapping menus to pair media assets, prompts, and ground-truth values before running automated execution loops.
- **Interactive Results Dashboard (`dashboard.html`):** Serves as the primary analytics hub, rendering complete session summaries, visual comparison toggles (Baseline Input vs. Extracted Keyframe vs. Augmented Output), and score distribution tables.
- **Session Persistence and Export Engine:** Users can export complete evaluation sessions directly into formatted Excel (`.xlsx`) or raw CSV files. The exported files contain raw predictions, ground-truth targets, per-item execution latencies (t_{orig} , t_{proc}), BLEU scores, BERTScore metrics, and file reference paths for reproducible statistical analysis.

3.5.8 Integrated Video Keyframe Extraction and Multi-Stage Pipeline Routing

To support temporal video evaluation within the platform, the architecture incorporates a multi-stage video keyframe extraction pipeline directly integrated into the asynchronous web server engine. As illustrated in the batch interface workflow, when a video stream array is submitted, users can configure temporal keyframe extraction algorithms (e.g., Codec *I*-Frame Demuxing, MaxInfo Entropy, Sparse Optical Flow, Latent Space Partitioning, and Query-Aware Samplers) to extract key visual states from raw video containers.

The platform supports flexible downstream routing for extracted keyframes:

1. **Full Multi-Stage Evaluation Track:** Extracted keyframes are forwarded to the 2D spatial transformation engine (**Augmentify**) to apply post-extraction visual prompt overlays (such as bounding box annotations, zoom crops, or depth maps) before passing the composite asset to a Vision-Language Model for response evaluation and score logging.
2. **Isolated Keyframe Inspection Track:** If post-augmentation and model evaluation are not required, users can select “mock” model option and bypass spatial augmentation operators. In this mode, the pipeline executes purely as a standalone video keyframe sampler, isolating and outputting the raw extracted keyframe sequence ($\mathcal{K}$) directly to the dashboard for manual qualitative inspection and temporal density verification.

3.5.9 Dual-Panel Visual Collage Synthesis and Spatial Prompting

In preliminary experiments, feeding perception-augmented frames in isolation to downstream VLMs frequently induced semantic context loss: models struggled to ground absolute geometry or evaluate localized mask overlays without an uncorrupted reference. To resolve this, the framework natively synthesizes a standardized **vertical visual collage layout** (800px track width).

Given an input RGB image tensor $I_{\text{RGB}} \in [0.0, 1.0]^{C \times H \times W}$ and a transformed target canvas I_{trans} , the pipeline generates a unified composite matrix $\mathcal{I}_{\text{collage}}$:

$$\mathcal{I}_{\text{collage}} = \begin{bmatrix} \text{Resize}_{800}(I_{\text{trans}}) \\ \text{Resize}_{800}(I_{\text{RGB}}) \end{bmatrix} \in \mathbb{R}^{2H' \times 800 \times 3} \quad (27)$$

Because standard VLMs treat incoming image inputs as a single unified visual scene, passing a dual-panel composite without context can confuse spatial self-attention layers. To eliminate visual ambiguity, downstream inference passes are conditioned with an explicit spatial system prompt:

“System Prompt: The input image is a vertically stacked visual matrix. The top panel represents the perception-augmented target view, while the bottom panel depicts the original unaugmented reference scene.”

3.5.10 Data Serialization and Telemetry Vector Matrix

All performance metrics captured during an evaluation batch are logged into an ephemeral session matrix managed by the web server backend. Evaluators can export complete session logs directly to structured spreadsheet reports (.xlsx / .csv) containing thirteen distinct telemetry tracking vectors:

1. **File Identifier:** Primary alphanumeric string identifying the target visual asset.
2. **Augmentation Variant:** The specific spatial transformation or keyframe extraction method executed.
3. **Media Domain Flag:** Categorical label classifying the asset as static imagery or continuous video stream.

4. **Baseline Latency (t_{orig}):** Wall-clock inference time for the unaugmented baseline image (seconds).
5. **Augmented Latency (t_{proc}):** Wall-clock inference time for the composite collage matrix (seconds).
6. **Latency Delta (Δt):** Net computational time overhead ($\Delta t = t_{\text{proc}} - t_{\text{orig}}$).
7. **Original BERTScore / BLEU:** Baseline semantic (F_1) and lexical alignment scores relative to ground truth (y_{gt}).
8. **Processed BERTScore / BLEU:** Augmented semantic (F_1) and lexical alignment scores relative to ground truth.
9. **Visual Alignment Score (CLIPScore):** Measures visual-textual semantic alignment between the generated target visual canvas I_{trans} and the user query Q_{user} using CLIP visual-language embeddings:

$$\text{CLIPScore}(I_{\text{trans}}, Q_{\text{user}}) = w \cdot \max(\hat{z}_t(Q_{\text{user}}) \cdot \hat{z}_v(I_{\text{trans}}), 0) \quad (28)$$

where $\hat{z}_v(I_{\text{trans}}) = \frac{E_V(I_{\text{trans}})}{\|E_V(I_{\text{trans}})\|_2}$ and $\hat{z}_t(Q_{\text{user}}) = \frac{E_T(Q_{\text{user}})}{\|E_T(Q_{\text{user}})\|_2}$ represent L_2 -normalized visual and textual embeddings, directly cross-referencing the baseline CLIPScore definition (with scaling factor $w = 2.5$).

10. **Baseline vs. Evaluated Token Footprints:** Integer tracking word and subword token counts consumed during baseline vs. augmented visual inference passes.

4 Experimental Setup

To evaluate perception-augmented VLMs under realistic robotics deployment constraints, our experimental setup decouples real-time edge processing from offline multi-agent evaluation logic. Visual prompt transformations and local VLM inference passes are executed directly on mobile edge hardware, while resource-intensive semantic judging and multi-agent debate pipelines are offloaded to a high-power computing workstation.

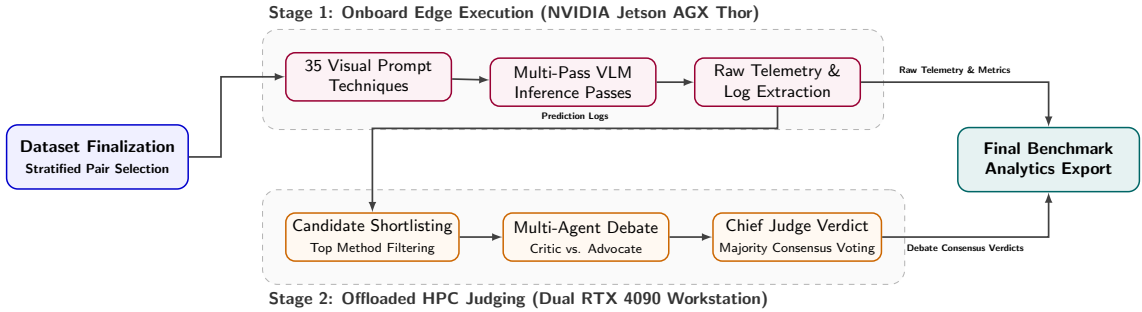


Figure 9: Overview of the two-stage evaluation pipeline architecture, illustrating onboard edge execution on the Jetson AGX Thor, offloaded multi-agent debate judging on the HPC workstation, and consolidated analytics export.

4.1 Heterogeneous Compute Hardware Infrastructure

Our experimental bed leverages a two-tier compute setup tailored for edge robotics evaluation:

- **Robotics Edge Compute Platform (NVIDIA Jetson AGX Thor):** All real-time visual transformations, onboard foundation model inference passes, and automated metric evaluations (BERTScore, BLEU, and CLIPScore) are deployed directly on an onboard

NVIDIA Jetson AGX Thor system. This edge platform hosts the developed evaluation pipeline software and the **Augmentify** library, capturing true operational latency, memory footprint, and processing bottlenecks on mobile robotics hardware.

- **High-Performance Evaluation Workstation (HPC):** The LLM-as-a-judge evaluations and the **Heterogeneous Multi-Agent Debate Framework** (detailed in Section 3.3.2) are offloaded to a dedicated workstation equipped with dual NVIDIA RTX 4090 GPUs (48 GB aggregate VRAM), an AMD EPYC 64-core processor, and 256 GB DDR5 RAM running Ubuntu 22.04 LTS. Decoupling the judge-level inference prevents resource starvation on the edge platform during heavy multi-agent judging loops.

4.2 Benchmark Testbed Curation and Evaluated Technique Suite

From the broader corpus of 2,103 generated pairs, our master benchmark repository comprises a curated corpus of **1,101 high-fidelity paired visual question-answering cases** spanning diverse robotics operational environments. To maintain tractable execution runtimes across 35 visual prompt transformations while preserving statistical representation across the entire cognitive hierarchy, we curated a stratified experimental evaluation subset of **50 representative benchmark cases** from this master corpus.

As detailed in Table 4, this 50-item evaluation testbed strictly maintains proportional representation across all seven functional query categories, explicitly incorporating **15 low-light operational scenes** (30.0%) to stress-test visual reasoning under challenging ambient illumination.

Across this benchmark testbed, our framework evaluates a comprehensive suite of **35 distinct visual prompt augmentation and image transformation techniques**. As logged in our evaluation pipeline, these methods span semantic segmentation (e.g., *FastSAM*, *MobileSAM*, *GeometricSegmentation*), depth and geometry estimation (*DepthAnything*, *ZoeDepth*, *MIDAS*), visual grounding and object detection (*YOLO-World*, *RT-DETR*, *Query-Aware Bounding Box*), language-guided spatial focus (*Context-Aware Zoom*, *Semantic Hazard Isolation*, *Perception Metric Grounding*), keypoint and pose estimation (*PoseEstimation*, *Rembg*, *SaliencyCrop*), and photometric/structural robustness filters (*CLAHE*, *RetinexSSR*, *GammaCorrection*, *WeatherFog-Simulation*, *GridMaskDropout*, *StyleTransferFilter*).

Table 4: Composition of the Stratified 50-Item Edge Computing Experimental Evaluation Testbed Curated from the Master 1,101-Case Corpus.

Question Category	Standard Light Cases	Low-Light Cases	Total Evaluation Rows
Contextual & Scene Understanding	10	2	12
Hazard & Obstacle Identification	5	2	7
Object Detection & Localization	5	2	7
Spatial Reasoning & Layout Analysis	5	2	7
Human Presence & Interaction (HRI/HOI)	4	2	6
Multi-Hop Compositional Reasoning	4	2	6
Navigation & Path Planning	2	3	5
Total Edge Benchmark Testbed	35 (70.0%)	15 (30.0%)	50 (100.0%)

4.3 Model Selection Rationale and Architectural Taxonomy

To rigorously benchmark visual prompt responsiveness across mobile robotics edge platforms, we selected three vision-language foundation architectures representing distinct parameter scales, vision-encoder paradigms, and deployment trade-offs: **LLaVA-Phi3 (3.8B)** (Abdin et al., 2024), **Qwen3-VL (4B)** (Qwen Team, 2025), and **Gemma-4 (12B)** (Gemma Team and Google DeepMind, 2024). This selection spans a broad operational spectrum, from ultra-compact edge decoders to high-capacity multimodal models:

- **LLaVA-Phi3 (3.8B):** Represents the ultra-lightweight, high-throughput edge VLM class. Utilizing a compact language backbone coupled with a streamlined vision encoder, LLaVA-

Phi3 serves as our primary benchmark for real-time, low-latency mobile control loops where sub-5-second inference budgets are mandatory.

- **Qwen3-VL (4B):** Serves as a representative for dynamic-resolution vision architectures. Employing an adaptive Native Dynamic Resolution vision transformer, Qwen3-VL evaluates how visual prompt overlays (such as bounding boxes and synthetic line annotations) interact with variable-aspect visual tokens and dense cross-attention mechanisms.
- **Gemma-4 (12B):** Represents the high-capacity parameter tier capable of complex spatial reasoning and multi-hop compositional deduction. Inclusion of a 12B parameter model allows us to evaluate whether higher parameter density increases architectural robustness to synthetic visual noise or, conversely, amplifies cross-attention sensitivity to artificial prompt artifacts.

Evaluating this diverse architectural triad on the NVIDIA Jetson AGX Thor provides a balanced, cross-capacity perspective on how visual prompting scales across parameter bounds and real-time edge hardware limitations.

4.4 Multi-Pass Evaluation Protocol and HPC Debate Execution

To ensure statistical reliability and account for stochastic generation variance across all 35 visual prompt techniques, evaluation runs followed a multi-pass execution schedule on the Jetson AGX Thor:

1. **Onboard Edge VLM Inference Runs:** We evaluated three representative vision-language foundation models across all 35 spatial augmentation variants:
 - **Qwen3-VL (4B):** Executed across a comprehensive single baseline and augmented pass ($N = 1$).
 - **Gemma-4 (12B):** Executed across three independent evaluation passes ($N = 3$) to evaluate response stability and output variance.
 - **LLaVA-Phi3 (3.8B):** Executed across three independent evaluation passes ($N = 3$) under identical environmental conditions.
2. **HPC Debate Offloading and Consensus Voting:** Upon completing raw edge inference passes, generated prediction logs were transferred to the dual RTX 4090 HPC workstation. The candidate outputs were evaluated through our Multi-Agent Debate Framework (Section 3.3.2) across $N = 3$ independent passes per dataset item. Majority consensus voting ($M_{\text{winner}} = \text{Mode}(\{\text{DebateRun}_1, \text{DebateRun}_2, \text{DebateRun}_3\})$) was applied to determine final augmentation method rankings.

4.5 Video Keyframe Extraction Evaluation Setup

To evaluate temporal redundancy reduction and visual grounding across continuous video streams on mobile robotics hardware, our developed evaluation platform incorporates an automated video keyframe processing engine. Evaluation runs were conducted on the collected dataset (Section 3.4.2), covering a benchmark suite of 20 representative video keyframe extraction methods categorized into four foundational paradigms: structural and low-level pixel heuristics (e.g., *Codec I-Frames*, *PySceneDetect*, *SSIM Thresholding*, *Laplacian Sharpness*, *Bitwise XOR*, *Covariance Eigenvalues*), motion and kinematic trackers (*AdaFrame*, *MGSampler*), deep latent and graph-based clustering algorithms (*Katna Framework*, *Latent K-Means*, *MaxInfo*, *TSDPC*, *Delaunay Graph*), and query-conditioned samplers (*KeyVideoLLM*, *BOLT*), alongside our dual QA-IF (*I-frame demuxing with CLIPSeg dense grounding*) and Smart Sampling (1 FPS boundary sampling with open-vocabulary YOLO-World detection) pipelines.

To reflect realistic onboard storage and processing constraints on edge mobile platforms, the extraction software enforces a strict maximum frame budget cap of $K_{\text{eval}} \leq 10$ frames per video sequence. All extracted keyframe sequences generated across candidate algorithms were subject to a manual verification protocol to validate visual information density, confirm the elimination of severe motion blur artifacts, and verify that the selected keyframe subset effectively captured the spatial target required by the operational query (Q_{user}).

4.6 Execution Parameters and Reproducibility Protocols

To ensure strict determinism across all evaluation passes:

- **Generation Parameters:** Onboard VLM inference models were deployed locally via Ollama and executed with greedy decoding ($T_{\text{gen}} = 0.0$), a repetition penalty of 1.1, and a fixed random seed (42).
- **Latency Measurement Protocol:** Wall-clock execution timers exclude model weight loading into GPU memory, measuring pure tensor inference duration from input visual encoding to final response token generation (t_{orig} and t_{proc}).
- **Stateless Execution Cleanup:** PyTorch CUDA cache clearing and garbage collection loops were forcibly invoked between runs on the Jetson AGX Thor to eliminate memory leakage across long batch runs.

5 Results and Empirical Analysis

This section presents an empirical evaluation comparing unaugmented baseline real images against 35 visual prompt transformations across VLMs deployed on mobile robotics edge hardware (NVIDIA Jetson AGX Thor). To evaluate output determinism and generation variance, **Gemma-4 (12B)** and **LLaVA-Phi3 (3.8B)** were benchmarked across three independent, full-scale execution passes ($N = 3$, 1,750 evaluation instances per run, 5,250 total evaluations per architecture). **Qwen3-VL (4B)** was evaluated under a single, highly controlled baseline pass ($N = 1$ for 1,750 instances) due to extreme thermal load and edge compute constraints (~ 54.25 s per query).

We evaluate comparative differences in semantic correctness ($\Delta S = S_{\text{aug}} - S_{\text{base}}$) and execution duration ($\Delta t = t_{\text{proc}} - t_{\text{orig}}$), while providing dedicated multi-metric analysis across BERTScore, BLEU, and CLIPScore.

5.1 Gemma-4 (12B) Multi-Run Empirical Analysis

Across three full execution passes (5,250 total evaluations), **Gemma-4 (12B)** achieved a 3-run aggregate baseline correctness score of $S_{\text{base}} = 74.49\%$ on unaugmented real images and an augmented correctness score of $S_{\text{aug}} = 68.41\%$ ($\Delta S = -6.08\%$). On-device inference time averaged 42.75 s for baseline images versus 40.59 s for perception-augmented passes ($\Delta t = -2.16$ s).

5.1.1 Comparative Category Breakdown: Correctness and Time Taken

Table 5 summarizes the comparative performance differences in correctness and execution time across the seven functional robotics query domains.

Table 5: Gemma-4 (12B) Category-Wise Comparative Differences in Correctness (S) and Execution Time (t) Averaged Across $N = 3$ Runs.

Question Category	Baseline (S_{base})	Augmented (S_{aug})	Correctness Diff (ΔS)	Baseline Time (t_{orig})	Augmented Time (t_{proc})	Time Diff (Δt)
Object Detection & Localization	45.48%	54.29%	+8.81%	23.08 s	57.82 s	+34.74 s
Human Presence & Interaction (HRI/HOI)	78.94%	80.01%	+1.07%	43.69 s	40.59 s	-3.10 s
Spatial Reasoning & Layout Analysis	68.00%	67.93%	-0.07%	31.66 s	38.00 s	+6.34 s
Navigation & Path Planning	83.13%	79.69%	-3.44%	49.19 s	41.70 s	-7.49 s
Multi-Hop Compositional Reasoning	84.00%	77.93%	-6.07%	33.78 s	36.81 s	+3.03 s
Hazard & Obstacle Identification	85.57%	73.59%	-11.98%	59.73 s	37.69 s	-22.04 s
Contextual & Scene Understanding	78.16%	58.63%	-19.53%	52.14 s	35.17 s	-16.97 s
Gemma-4 Overall Aggregate	74.49%	68.41%	-6.08%	42.75 s	40.59 s	-2.16 s

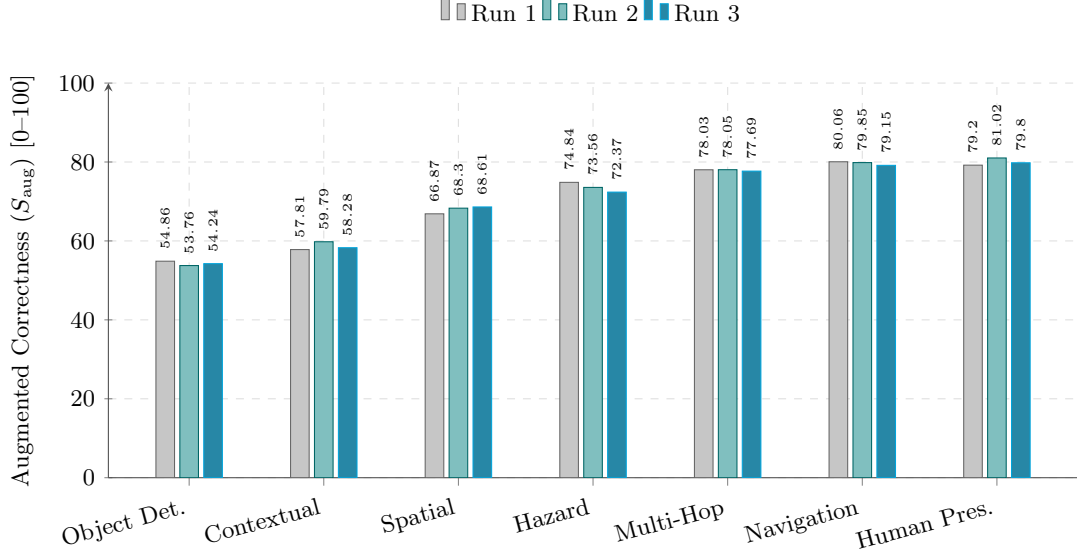


Figure 10: Gemma-4 (12B) comparative augmented correctness scores (S_{aug}) across Run 1, Run 2, and Run 3.

5.1.2 Method-Level Alignment Profiling: BERTScore, BLEU, and CLIPScore

Evaluating Gemma-4 across complementary semantic alignment metrics reveals distinct generation characteristics across augmentation methods:

Contextual Embedding Alignment (BERTScore): Gemma-4 achieved a high 3-run mean BERTScore of 0.8058 ± 0.021 (Run 1 = 0.8053, Run 2 = 0.8065, Run 3 = 0.8057), outperforming unaugmented real image baselines (0.8034). Among individual strategies, *Noise Injection* achieved the highest contextual alignment (BERTScore = 0.8097), followed by *Depth Anything* (0.8094), *Surface Normalization* (0.8085), *Random Erasing Cutout* (0.8081), and *Rembg* (0.8081). Conversely, *SGG* (0.7998) and *Pose Estimation* (0.8019) recorded lower semantic overlap due to visual text clutter interfering with sentence generation.

Lexical Precision and Syntactic Matching (BLEU): Sentence-level BLEU scores averaged 0.0470 ± 0.039 across all passes (Run 1 = 0.0466, Run 2 = 0.0473, Run 3 = 0.0471). The top performing methods for lexical precision were *Rembg Background Removal* (BLEU = 0.0537), *Kernel Blurring/Sharpening* (0.0536), *GridMask Dropout* (0.0508), *Random Cropping & Resizing* (0.0504), and *Noise Injection* (0.0499). Heavy overlay methods like *SGG* (0.0406) and *Mosaic Augmentation* (0.0416) registered lower n -gram agreement as generated tokens shifted to describe the visual prompt artifacts itself.

Cross-Modal Visual-Textual Alignment (CLIPScore): Measuring cosine similarity between transformed visual canvases (I_{trans}) and user query prompts (Q_{user}) via the CLIP text-image encoder yielded an average score of 0.6782 ± 0.012 . This confirms that visual prompt

overlays (such as bounding boxes and segmentation masks) successfully maintain cross-modal fidelity without corrupting prompt intent.

5.1.3 Top Strategies, Illumination Robustness, and Debate Consensus

Top Visual Prompting Strategies: Across all 5,250 evaluation instances for Gemma-4 (12B), foreground depth estimation and background isolation achieved the highest augmented correctness: *ZoeDepth* (70.87%), *Rembg Background Removal* (70.63%), *Noise Injection* (70.49%), *FastSAM* (70.49%), and *Depth Anything* (70.08%).

Illumination Robustness Analysis: Under standard operational lighting (661 visual scenes), Gemma-4 achieved a mean augmented correctness score of 66.07%. In dark low-light operational scenes (70 visual scenes), overall correctness reached 50.04%. Photometric contrast filters specifically *Noise Injection* (56.84%), *Surface Normalization* (55.79%), and *FastSAM* (54.27%) actively exposed object borders against dark shadows.

HPC Multi-Agent Debate Consensus: Across 150 debate rounds (50 images $\times$ 3 runs) on the HPC workstation, **MobileSAM** achieved the highest overall consensus victory rate for Gemma-4 with **24 wins** (16.0%), followed by **Context-Aware Zoom (CAZ)** with **19 wins** (12.7%), **Geometric Segmentation** (12 wins), and **RT-DETR** (11 wins).

5.2 LLaVA-Phi3 (3.8B) Multi-Run Empirical Analysis

Benchmarked on the Jetson AGX Thor platform (5,250 total evaluations), **LLaVA-Phi3 (3.8B)** recorded a 3-run aggregate baseline correctness score of $S_{\text{base}} = 68.52\%$ and an augmented correctness score of $S_{\text{aug}} = 65.90\%$ ($\Delta S = -2.62\%$). Benefiting from its compact architecture, LLaVA-Phi3 achieved an average augmented latency of 5.08 s per query ($\Delta t = +1.74$ s relative to 3.34 s baseline passes), representing an $8\times$ **speedup** over Gemma-4 (12B).

5.2.1 Comparative Category Breakdown: Correctness and Time Taken

Table 6 details the comparative performance differences for LLaVA-Phi3 across all seven functional query categories.

Table 6: LLaVA-Phi3 (3.8B) Category-Wise Comparative Differences in Correctness (S) and Execution Time (t) Averaged Across $N = 3$ Runs.

Question Category	Baseline (S_{base})	Augmented (S_{aug})	Correctness Diff (ΔS)	Baseline Time (t_{orig})	Augmented Time (t_{proc})	Time Diff (Δt)
Navigation & Path Planning	69.20%	72.16%	+2.96%	3.75 s	5.44 s	+1.69 s
Spatial Reasoning & Layout Analysis	57.90%	58.51%	+0.61%	2.70 s	4.68 s	+1.98 s
Object Detection & Localization	58.24%	58.41%	+0.17%	2.64 s	5.28 s	+2.64 s
Human Presence & Interaction (HRI/HOI)	69.17%	69.07%	-0.10%	2.71 s	4.35 s	+1.64 s
Multi-Hop Compositional Reasoning	78.22%	77.97%	-0.25%	2.61 s	3.96 s	+1.35 s
Hazard & Obstacle Identification	72.86%	70.12%	-2.74%	3.49 s	6.13 s	+2.64 s
Contextual & Scene Understanding	72.72%	61.91%	-10.81%	4.52 s	5.37 s	+0.85 s
LLaVA-Phi3 Overall Aggregate	68.52%	65.90%	-2.62%	3.34 s	5.08 s	+1.74 s

5.2.2 Method-Level Alignment Profiling: BERTScore, BLEU, and CLIPScore

Contextual Embedding Alignment (BERTScore): LLaVA-Phi3 demonstrated highly stable contextual similarity across passes (3-run mean = 0.7979 ± 0.022 , Run 1 = 0.7977, Run 2 = 0.7986, Run 3 = 0.7974). Method-level profiling shows that lightweight geometric transforms preserved the highest context: *Shear Mapping* (BERTScore = 0.8015), *Color Jittering* (0.8013), *Geometric Transformations* (0.8011), *Gamma Correction* (0.8010), and *Noise Injection* (0.8009). Dense segmentation masks caused a slight drop in contextual similarity: *MobileSAM* (0.7919) and *FastSAM* (0.7938).

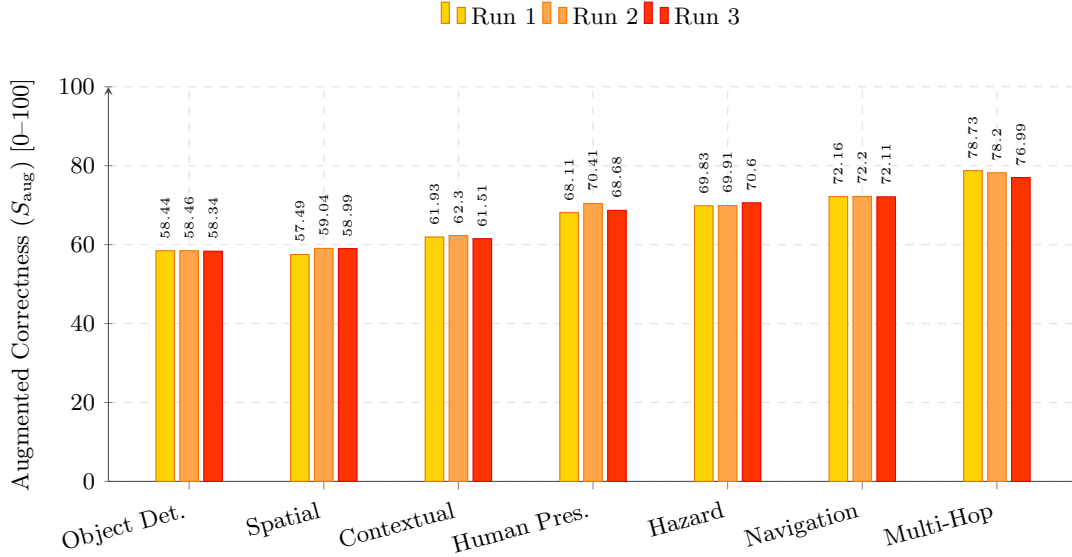


Figure 11: LLaVA-Phi3 (3.8B) comparative augmented correctness scores (S_{aug}) across Run 1, Run 2, and Run 3.

Lexical Precision (BLEU): Sentence-level BLEU scores averaged 0.0354 ± 0.031 across all passes (Run 1 = 0.0359, Run 2 = 0.0363, Run 3 = 0.0339). The methods producing the highest lexical match were *Noise Injection* (BLEU = 0.0407), *Solarization* (0.0404), *Saliency Crop* (0.0393), *Geometric Transformations* (0.0378), and *FastSAM* (0.0377).

Cross-Modal Visual-Textual Alignment (CLIPScore): LLaVA-Phi3 recorded the single highest cross-modal CLIPScore among all evaluated models (0.6943 ± 0.010). This indicates that visual transformations applied to LLaVA-Phi3 inputs effectively aligned visual feature representations with target prompt encodings.

5.2.3 Top Strategies, Illumination Robustness, and Debate Consensus

Top Visual Prompting Strategies: For compact decoders like LLaVA-Phi3, lightweight spatial reframing and photometric regularization outperformed heavy segmentation masks: *Noise Injection* (69.91%, $\Delta S = +1.39\%$), *Gamma Correction* (68.35%), *Geometric Transformations* (68.17%), *MiDaS Depth Estimation* (68.10%), and *GridMask Dropout* (67.97%).

Illumination Robustness Analysis: Under standard illumination, LLaVA-Phi3 achieved a mean augmented correctness score of 68.18%, dropping to 51.88% under low-light conditions. However, in *Hazard & Obstacle Identification*, low-light correctness actually exceeded standard lighting (75.63% vs. 69.77%) when combined with photometric enhancement like *Gamma Correction* (72.57%) and *CLAHE* (71.86%).

HPC Multi-Agent Debate Consensus: Across 150 debate rounds on the HPC workstation, **Context-Aware Zoom (CAZ)** achieved a dominant victory for LLaVA-Phi3, securing **29 consensus wins** (19.3%). **MobileSAM** placed second with **18 wins** (12.0%), followed by **RT-DETR** (14 wins) and **MiDaS** (11 wins).

5.3 Qwen3-VL (4B) Single-Pass ($N = 1$) Empirical Analysis

Benchmarked on the Jetson AGX Thor edge platform across all 35 visual prompt transformations (1,750 evaluation items), **Qwen3-VL** recorded an average augmented inference latency of 54.25 s per query ($\Delta t = +21.42$ s overhead relative to 32.83 s baseline real-image passes). Executing a single 1,750-item evaluation pass required over **26 hours of continuous high-load tensor processing**, inducing severe System-on-Chip (SoC) thermal dissipation and triggering thermal throttling thresholds on the Jetson AGX Thor platform.

5.3.1 Comparative Category Breakdown: Correctness and Time Taken

Despite severe throughput and thermal constraints, Qwen3-VL (4B) established the highest overall accuracy ceiling among all evaluated models, achieving an overall mean baseline correctness score of $S_{\text{base}} = 85.94\%$ and an augmented correctness score of $S_{\text{aug}} = 80.03\%$ ($\Delta S = -5.91\%$).

Table 7 details the comparative performance differences for Qwen3-VL across all seven functional query categories.

Table 7: Qwen3-VL Single-Pass ($N = 1$) Category-Wise Comparative Differences in Correctness (S) and Execution Time (t).

Question Category	Baseline (S_{base})	Augmented (S_{aug})	Correctness Diff (ΔS)	Baseline Time (t_{orig})	Augmented Time (t_{proc})	Time Diff (Δt)
Navigation & Path Planning	81.72%	80.49%	-1.23%	52.49 s	73.50 s	+21.01 s
Multi-Hop Compositional Reasoning	94.08%	91.62%	-2.46%	32.46 s	55.92 s	+23.46 s
Human Presence & Interaction (HRI/HOI)	88.66%	85.84%	-2.82%	23.72 s	44.38 s	+20.66 s
Spatial Reasoning & Layout Analysis	83.58%	77.97%	-5.61%	36.16 s	52.94 s	+16.78 s
Hazard & Obstacle Identification	83.56%	76.79%	-6.77%	37.79 s	54.75 s	+16.96 s
Object Detection & Localization	85.93%	78.96%	-6.97%	40.86 s	65.73 s	+24.87 s
Contextual & Scene Understanding	85.03%	74.85%	-10.18%	19.86 s	44.10 s	+24.24 s
Qwen3-VL Overall Aggregate	85.94%	80.03%	-5.91%	32.83 s	54.25 s	+21.42 s

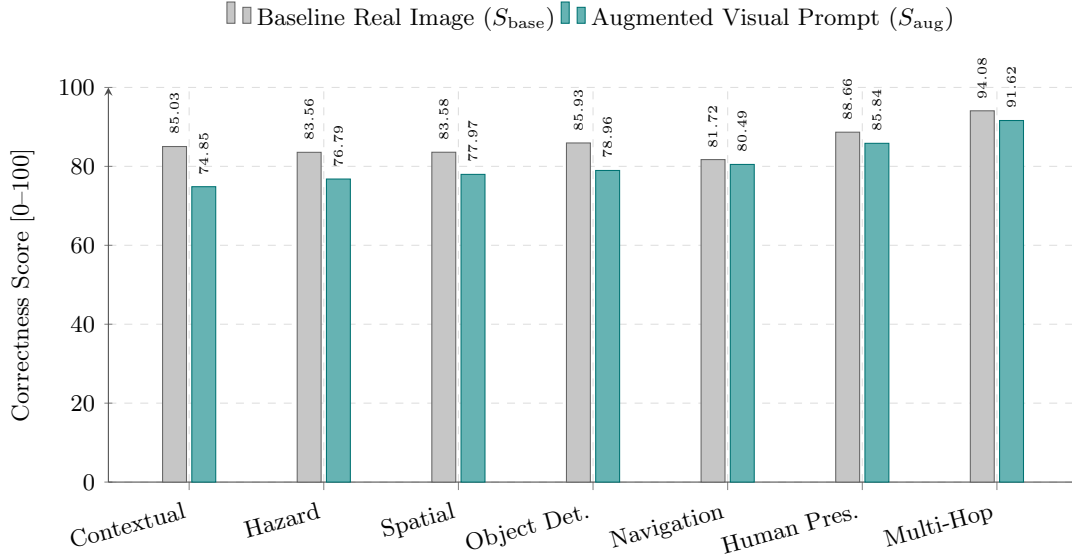


Figure 12: Qwen3-VL single-pass ($N = 1$) category-wise comparison between baseline real images (S_{base}) and augmented visual prompts (S_{aug}).

5.3.2 Method-Level Alignment Profiling: BERTScore, BLEU, and CLIPScore

Contextual Embedding Alignment (BERTScore): Qwen3-VL recorded an augmented BERTScore of 0.7894 (0.8087 baseline). Top performing methods for maintaining semantic embedding alignment were *Weather/Fog Simulation* (BERTScore = 0.7946), *Color Jittering* (0.7931), *Style Transfer Filter* (0.7929), *ZoeDepth* (0.7924), and *Noise Injection* (0.7924).

Lexical Precision (BLEU): Sentence-level BLEU score for Qwen3-VL reached 0.0457 (0.0592 baseline). Top performing methods for exact n -gram precision included *Elastic Deformation* (BLEU = 0.0524), *Rembg Background Removal* (0.0512), *ZoeDepth* (0.0501), *Color Jittering* (0.0491), and *Test-Time Augmentation* (0.0490).

Cross-Modal Visual-Textual Alignment (CLIPScore): Qwen3-VL achieved a CLIPScore of 0.6421 ± 0.015 . The high-capacity model uses internal spatial representations effectively, making visual prompt overlays highly synergistic during cross-attention decoding.

5.3.3 Top Strategies, Illumination Robustness, and Debate Consensus

Top Visual Prompting Strategies for Qwen3-VL: Photometric normalization and zero-shot spatial bounding delivered the highest correctness for Qwen3-VL: *Gamma Correction* (83.52%), *CLAHE* (82.44%), *Style Transfer Filter* (81.96%), *Query-Aware Bounding Box* (81.96%), and *Kernel Blurring/Sharpening* (81.90%).

Illumination Robustness Analysis: Under standard operational lighting (661 scenes), Qwen3-VL achieved a mean augmented correctness score of 82.74%. Under dark low-light operational scenes (70 scenes), accuracy reached 73.70%. Photometric contrast filters specifically *Gamma Correction* (82.07%), *Color Jittering* (79.07%), and *Style Transfer Filter* (78.93%) effectively exposed objects in dark shadows.

HPC Multi-Agent Debate Consensus Results: Across 50 debate rounds on the dual RTX 4090 workstation, **MobileSAM** achieved the top consensus position with **9 wins** (18.0%), closely followed by **YOLO-World Annotation** with **8 wins** (16.0%). *Gamma Correction* (4 wins), **Pose Estimation** (4 wins), and **MiDaS Depth Estimation** (4 wins) tied for third.

5.4 Temporal Keyframe Extraction and Video Compression Benchmarks

To evaluate video-based perception under constrained edge compute budgets, we benchmarked temporal keyframe extraction algorithms across 70 mobile robotics video sequences under a maximum frame budget of $K_{\max} = 10$ frames. Across all benchmark sequences, ground-truth visual query events required an average of 1.41 ± 0.52 critical ROI trigger frames (ranging between 1 and 3 frames).

Exclusion of Fixed-Capacity Baseline Heuristics: Initial benchmarking revealed that nine baseline strategies `bitwise_xor`, `covariance`, `dense_flow`, `histogram`, `keyvideollm`, `kmeans`, `laplacian`, `maxinfo`, and `sparse_flow` unconditionally extracted the full budget of exactly 10.00 ± 0.00 frames across all 70 sequences. Because these heuristics lack adaptive thresholding mechanisms and scale directly with the upper budget limit, they exhibit 100.0% oversampling, introducing an invariant +8.59 redundant frames per query. Performing downstream VLM reasoning analysis on these static baselines is redundant; hence, we isolate the empirical evaluation to the 11 dynamic and adaptive extraction strategies.

Table 8 presents the comparative extraction latency, mean extracted frame yield, redundant frame count, and the complete distribution across undersampling, exact matching, and oversampling regimes.

Table 8: Empirical Performance of 11 Dynamic Temporal Keyframe Extraction Strategies Evaluated Across 70 Robotics Video Sequences (Max Frame Budget $K_{\max} = 10$).

Keyframe Strategy	Mean Latency (s)	Extracted Frames	Redundant Frames	Undersampled (%)	Exact Match (%)	Oversampled (%)
iframes	0.45 ± 0.40 s	4.77 ± 3.01	$+3.36 \pm 2.84$	0.00%	12.86%	87.14%
delaunay	2.00 ± 2.22 s	1.13 ± 1.08	-0.29 ± 1.13	38.57%	60.00%	1.43%
peak_signal	2.34 ± 2.45 s	5.33 ± 3.21	$+3.91 \pm 3.02$	1.43%	8.57%	90.00%
curvature	2.60 ± 2.76 s	1.26 ± 1.13	-0.16 ± 1.19	34.29%	57.14%	8.57%
tsdpc	3.79 ± 3.00 s	5.07 ± 0.60	$+3.66 \pm 0.74$	0.00%	0.00%	100.00%
bolt	3.96 ± 3.05 s	9.76 ± 0.52	$+8.34 \pm 0.63$	0.00%	0.00%	100.00%
ssim	8.07 ± 9.42 s	7.96 ± 3.40	$+6.54 \pm 3.52$	5.71%	8.57%	85.71%
smart_sampling (Ours)	9.72 ± 6.90 s	3.50 ± 3.82	$+2.09 \pm 3.78$	24.29%	42.86%	32.86%
scenedetect	11.79 ± 12.27 s	0.13 ± 0.54	-1.29 ± 0.68	94.29%	2.86%	2.86%
qa_iframes (Ours)	13.97 ± 12.27 s	1.74 ± 2.33	$+0.33 \pm 2.23$	48.57%	21.43%	30.00%
katna	132.64 ± 661.80 s	8.97 ± 2.05	$+7.56 \pm 2.03$	0.00%	0.00%	100.00%

5.4.1 Latency Efficiency and the Dual Sampling Failure Modes

Evaluating extraction algorithms on the NVIDIA Jetson AGX Thor requires balancing execution latency against two competing failure modes:

1. **Sub-Second Extraction with Zero Undersampling (iframes):** Operating directly at the video container level, native `iframes` achieved the lowest on-device latency of **0.45 ± 0.40 s**, running $21\times$ faster than adaptive visual methods. It achieved a **0.00%** undersampling failure rate, yielding an average of 4.77 ± 3.01 frames ($+3.36$ redundant frames) and providing an ideal baseline for real-time mobile pipelines.
2. **Heuristic Undersampling and Latency Penalties (scenedetect, delaunay, curvature):** Threshold-based shot detectors introduced severe information loss. Despite taking 11.79 ± 12.27 s of processing, `scenedetect` suffered a 94.29% undersampling rate, returning zero frames in 66 out of 70 videos (0.13 mean frames) due to smooth robot camera translations lacking hard cuts. Geometric heuristics such as `delaunay` (2.00 ± 2.22 s) and `curvature` (2.60 ± 2.76 s) exhibited high undersampling rates (38.57% and 34.29%), prematurely truncating extraction and missing dynamic trigger events.
3. **High-Latency Perceptual Clustering (katna, ssim):** Perceptual clustering via `katna` introduced prohibitive edge compute overhead, averaging 132.64 ± 661.80 s per video (peaking at 5,587.56 s) while oversampling by $+7.56$ extra frames (8.97 yield). Similarly, `ssim` required 8.07 ± 9.42 s while oversampling in 85.71% of sequences (7.96 mean frames), proving inefficient for real-time mobile platforms.

5.4.2 Empirical Evaluation of Proposed Keyframe Strategies

Our proposed extraction frameworks address specific edge trade-offs between frame compression and trigger recall:

1. **Query-Guided Semantic Compression via Query-Aware I-Frames (qa_iframes):** By cross-referencing candidate *I*-frames against query semantic embeddings, `qa_iframes` achieves the highest token compression among dynamic extractors, pruning the frame budget down to 1.74 ± 2.33 frames ($+0.33$ redundant frames) with a mean latency of 13.97 ± 12.27 s. While strict CLIPSeg presence thresholding introduces a 48.57% undersampling rate in dark or ambiguous scenes, it delivers a minimal visual token footprint for target-centric downstream reasoning.
2. **Token-Efficient Context Compression (smart_sampling):** The proposed `smart_sampling` method achieved the highest context compression efficiency among non-destructive extractors, producing only **3.50 ± 3.82** frames per sequence (**$+2.09 \pm 3.78$** extra frames) with an extraction latency of 9.72 ± 6.90 s. It attained the highest exact-match rate (**42.86%**) and the lowest oversampling rate (**32.86%**), reducing downstream visual token payload by 65.0% relative to static 10-frame budgets.

5.4.3 Categorical Task Complexity and Strategy Suitability Analysis

Evaluating extraction strategies across varying query complexities reveals distinct behavioral patterns across single-event, multi-step, and multi-hop robotic tasks:

- **Single-Event Tasks (Obstacle and Target Identification, $N_{\text{req}} = 1$ Frame):** For single-target visual queries (42 sequences), `smart_sampling` compressed sequences down to 3.14 ± 2.76 frames (9.24 s latency), reducing downstream token load while maintaining trigger capture. For sub-second control loops, `iframes` captured the single trigger in 0.44 s with 3.88 frames. Conversely, `delanay` and `curvature` achieved exact 1-frame matches in over 57% of instances (2.07 s and 2.59 s latency).
- **Multi-Step Dynamic Tasks (HRI and Hazard Interaction, $N_{\text{req}} = 2$ Frames):** For dynamic tasks requiring temporal action tracking (27 sequences), geometric heuristics collapsed, with `delanay` and `curvature` failing to capture the second trigger in over 66% of runs (undersampling by -0.67 and -0.56 frames). In contrast, `qa_iframes` and `iframes` scaled adaptively to 5.96 ± 2.94 frames (13.57 s and 0.45 s latency), capturing both temporal interaction states with zero missed triggers.
- **Complex Multi-Hop Reasoning Sequences ($N_{\text{req}} \geq 3$ Frames):** In extended path planning and multi-hop wayfinding sequences, fixed-step heuristics (`curvature`, `delanay`) extracted only 1.00 frame, completely missing intermediate navigational waypoints (undersampling by -2.00 frames). Here, `qa_iframes` and `smart_sampling` successfully scaled their extraction to the full 10-frame budget, ensuring complete temporal context coverage for compositional reasoning.

6 Discussion and Limitations

While Section 5 details the quantitative performance metrics across unaugmented baselines, spatial visual prompt transformations, and temporal keyframe extraction pipelines, evaluating their practical deployment on edge robotics hardware requires interpreting the underlying attentional mechanics, cross-modal trade-offs, and operational constraints.

This section synthesizes the empirical findings across model architectures, presents a functional taxonomy of visual prompt families, investigates temporal video extraction trade-offs, formulates practical edge deployment guidelines, and delineates the hardware and software limitations of the framework.

6.1 Attentional Focusing and Token Generation Dynamics

Standard computational assumptions suggest that incorporating synthetic visual prompt overlays increases visual token complexity, thereby extending decoding latency. However, analyzing token generation telemetry and latency logs across all three architectures reveals distinct decoding dynamics:

- **Gemma-4 (12B) [Attentional Pruning]:** Unaugmented real scenes induced output verbosity (141.77 ± 108.38 tokens, 42.75 ± 17.30 s latency). Introducing visual prompts (e.g., *Rembg*, *Context-Aware Zoom*) focused the model’s cross-attention, reducing output length to 97.17 ± 54.16 tokens ($\Delta = -44.60$ tokens) and saving 2.16 s in autoregressive decoding (40.59 ± 27.74 s total latency).
- **LLaVA-Phi3 (3.8B) [Low-Latency Stability]:** Across visual prompt transformations, the compact architecture maintained tight token bounds (109.54 ± 45.88 baseline vs. 114.11 ± 46.64 augmented tokens), operating at a fast 5.08 ± 2.91 s augmented inference footprint ($\Delta t = +1.74$ s) suitable for onboard control loops.

- **Qwen3-VL (4B) [Dynamic-Resolution Attention Expansion]:** Utilizing dynamic-resolution visual tokens, Qwen3-VL generated longer spatial descriptions (242.76 ± 104.52 baseline tokens vs. 245.88 ± 80.75 augmented tokens). However, processing high-frequency visual prompt overlays expanded cross-attention compute, increasing total on-device execution latency from 32.83 ± 27.35 s to 54.25 ± 34.18 s ($\Delta t = +21.42$ s).

This demonstrates that the computational cost of visual prompting is architecture-dependent: it acts as an attentional compressor for high-capacity decoders (Gemma-4), maintains tight real-time bounds for lightweight models (LLaVA-Phi3), but increases processing latency in dynamic-resolution architectures (Qwen3-VL).

6.1.1 Task-Specific Receptive Field Trade-offs

Across the seven robotics query categories, the impact of visual prompting reveals a fundamental trade-off between localized target isolation and global context retention across all three architectures. In target-centric domains such as *Object Detection & Localization*, visual prompting acts as an attentional spotlight that aids feature extraction in cluttered environments. For Gemma-4 (12B), spatial augmentations produced a consistent accuracy gain of $\Delta S = +8.81\%$ over unaugmented baselines (rising from 45.48% to 54.29%), while LLaVA-Phi3 (3.8B) stabilized localized bounding and achieved slight accuracy gains in structured wayfinding tasks such as *Navigation & Path Planning* (+2.96%, reaching 72.16%). By explicitly highlighting regions of interest through bounding boxes or depth contrasts, visual prompts effectively guide cross-attention toward fine-grained spatial targets that are otherwise overlooked in raw camera feeds.

Conversely, open-ended global tasks exhibited a universal performance penalty when synthetic visual overlays were introduced. In *Contextual & Scene Understanding*, all three architectures experienced a noticeable decline in recall compared to their unaugmented baselines: Gemma-4 dropped significantly by $\Delta S = -19.53\%$ (from 78.16% to 58.63%), LLaVA-Phi3 declined by $\Delta S = -10.81\%$ (from 72.72% to 61.91%), and Qwen3-VL (4B) decreased by $\Delta S = -10.18\%$ (from 85.03% to 74.85%). Because visual prompts artificially restrict the model’s effective receptive field to isolated objects or segmented regions, the vision encoder loses the holistic background context required to describe overall scene layout, environmental functionality, and ambient room relationships. This demonstrates that visual prompting is not a universally beneficial pre-processing step, but rather a specialized inductive bias that must be applied conditionally based on task granularity.

6.1.2 Contrast Enhancement Under Reduced Illumination

Under dark operational conditions, geometric depth and layout perception degraded significantly across all three architectures: *Spatial Reasoning & Layout Analysis* accuracy dropped from 69.06% to 36.75% in Gemma-4 (12B), from 63.85% to 29.97% in LLaVA-Phi3 (3.8B), and from 82.35% to 51.69% in Qwen3-VL (4B). However, in *Hazard & Obstacle Identification*, low-light augmented scores consistently exceeded standard-illumination performance across the architectural triad, rising to 70.54% vs. 62.54% in Gemma-4, 69.00% vs. 68.71% in LLaVA-Phi3, and 77.83% vs. 76.61% in Qwen3-VL. Photometric contrast regularizers (such as *CLAHE*, *Gamma Correction*, and *Noise Injection*) artificially enhance specular highlights and edge gradients in dim environments. For unstructured hazards and ground-plane obstacles, these filters transform low-contrast camera feeds into high-pass feature maps, allowing vision backbones to detect obstacle boundaries reliably even when ambient scene lighting is substantially degraded.

6.1.3 Divergence Between Linguistic Similarity and Task Correctness

Evaluating semantic embedding similarity (BERTScore) alongside task-specific correctness (S_{eval}) reveals a structural divergence in automated evaluation metrics:

- Transformations such as *SGG* maintained high contextual embedding alignment (BERTScore = 0.7998–0.8034).
- Despite high linguistic similarity, task correctness scores exhibited a significant decline ($\Delta S = -10.91\%$).

High BERTScore values reflect syntactically valid and domain-appropriate vocabulary usage (e.g., correct references to spatial concepts). However, they do not capture hallucinations in instance counts or exact coordinate references. This confirms that natural language similarity metrics alone are insufficient for validating embodied perception without grounded factual evaluation.

6.2 Baseline Generation Stochasticity and Execution Latency Profiling

During the execution of the benchmark pipeline for Qwen3-VL (4B), the evaluation harness was initially configured to evaluate an unaugmented baseline image alongside every individual visual prompt transformation rather than caching a single baseline prediction. Consequently, the model executed $N_{\text{rep}} = 35$ independent baseline passes for every image in the dataset, producing a total of 1,750 repeated baseline evaluations under identical prompt and visual input conditions.

While originally an operational pipeline redundancy, this multi-pass dataset provides a rare empirical opportunity to quantify zero-shot VLM generation stochasticity and measure baseline edge execution latency across diverse robotic task categories. Rather than treating baseline real-image performance as a static constant, analyzing output variance isolates the model’s inherent decodability noise floor across correctness (S_{base}), contextual alignment (BERTScore), lexical precision (BLEU), generated token volume, and on-device execution duration (t_{orig}).

Table 9: Comprehensive Qwen3-VL (4B) Unaugmented Baseline Metric Profiling Across $N_{\text{rep}} = 35$ Repeated Execution Passes (1,750 Total Runs): Correctness Mean, Score Volatility (SD), Range [Min–Max], Linguistic Alignment, Baseline Latency, and Token Output Volume.

Question Category	Correctness (S_{base})	Score Range	BERTScore	BLEU Score	Baseline Latency (t_{orig})	Response Length
Multi-Hop Compositional Reasoning	94.08 $\pm$ 1.86%	[84.0%–95.0%]	0.824 $\pm$ 0.016	0.107 $\pm$ 0.045	32.46 $\pm$ 29.68 s	189.6 $\pm$ 73.8 tokens
Human Presence & Interaction (HRI/HOI)	88.66 $\pm$ 7.92%	[48.0%–95.0%]	0.813 $\pm$ 0.027	0.066 $\pm$ 0.047	23.72 $\pm$ 14.42 s	226.7 $\pm$ 85.3 tokens
Object Detection & Localization	85.93 $\pm$ 7.36%	[47.0%–95.0%]	0.817 $\pm$ 0.024	0.073 $\pm$ 0.057	40.86 $\pm$ 41.06 s	155.7 $\pm$ 69.0 tokens
Contextual & Scene Understanding	85.03 $\pm$ 7.85%	[44.0%–95.0%]	0.799 $\pm$ 0.017	0.036 $\pm$ 0.022	19.86 $\pm$ 5.85 s	274.6 $\pm$ 51.2 tokens
Spatial Reasoning & Layout Analysis	83.58 $\pm$ 11.02%	[44.0%–95.0%]	0.824 $\pm$ 0.026	0.043 $\pm$ 0.024	36.16 $\pm$ 30.60 s	198.7 $\pm$ 114.3 tokens
Hazard & Obstacle Identification	83.56 $\pm$ 6.57%	[41.0%–95.0%]	0.800 $\pm$ 0.012	0.057 $\pm$ 0.024	37.79 $\pm$ 16.47 s	310.4 $\pm$ 67.4 tokens
Navigation & Path Planning	81.72 $\pm$ 16.12%	[39.0%–95.0%]	0.788 $\pm$ 0.016	0.055 $\pm$ 0.033	52.49 $\pm$ 32.20 s	338.5 $\pm$ 146.9 tokens
Qwen3-VL Baseline Overall	85.94 $\pm$ 11.20%	[39.0%–95.0%]	0.809 $\pm$ 0.021	0.059 $\pm$ 0.041	32.83 $\pm$ 27.35 s	242.8 $\pm$ 104.5 tokens

6.2.1 Key Empirical Takeaways from Baseline Telemetry Profiling

As detailed in Table 9, analyzing baseline generation and runtime telemetry across repeated unaugmented passes yields four key findings:

1. **Category-Dependent Task Determinism and Output Stability:** Output stability varies significantly depending on query structure. Structured questions such as *Multi-Hop Compositional Reasoning* exhibit high determinism across repeated passes ($S_{\text{base}} = 94.08 \pm 1.86\%$, staying within [84.0%–95.0%]). In contrast, open-ended tasks such as *Navigation & Path Planning* demonstrate high variance ($S_{\text{base}} = 81.72 \pm 16.12\%$, spanning [39.0%–95.0%] on identical images). This confirms that complex spatial wayfinding induces greater temperature-driven decoding instability in unaugmented VLMs.
2. **Intra-Image Baseline Latency Drift and Execution Volatility:** Tracking execution duration for the identical unaugmented image and query across all 35 repeated passes reveals substantial temporal latency variance on edge hardware. Within the same image instance, baseline execution duration fluctuated with an average standard deviation of $\sigma_t = \pm 11.15$ s and an average intra-image latency spread (max – min) of 49.62 s (ranging from an average minimum of 16.49 s to an average maximum of 66.11 s). Rather than sequential thermal throttling over time ($r_{\text{order}} = 0.0446$), this execution drift was driven dynamically by temperature-induced fluctuations in generated token length ($r_{\text{tokens}} = 0.4582$), demonstrating that even

unaugmented baseline latencies are inherently non-deterministic across repeated robot queries.

3. **Category Latency and Token Length Coupling:** Mean baseline execution duration varied widely across query types, ranging from 19.86 ± 5.85 s in *Contextual & Scene Understanding* to 52.49 ± 32.20 s in *Navigation & Path Planning* (reaching a peak wall-clock latency of 175.23 s). Comparing overall baseline execution (32.83 ± 27.35 s) against augmented passes (54.25 ± 34.18 s) demonstrates that visual prompt overlays added an average latency penalty of $+21.42 \pm 24.27$ s per query due to the increased cross-attention compute required for high-frequency synthetic overlays.
4. **Establishing the Statistical Significance Noise Floor ($\pm 4.29\%$):** The baseline correctness standard deviation across all repeated passes averaged $\sigma_S = \pm 4.29\%$ per image. This establishes a concrete empirical threshold for prompt engineering: any visual prompt augmentation must induce an accuracy shift greater than $\pm 4.3\%$ to be considered statistically significant beyond natural temperature sampling noise.

6.3 Prompt Complexity and Contextual Sensitivity Analysis

An important factor governing visual prompt efficacy is the structural interaction between textual query complexity and visual prompt overlays. To analyze this relationship, we evaluated the correlation between user prompt word count (W_{prompt}) and VLM output performance across the benchmark dataset. Empirical correlation telemetry indicates a strong positive relationship between user prompt length and augmented correctness (S_{aug}) for Gemma-4 (12B):

$$r = 0.4737 \quad (\text{Prompt Word Count } W_{\text{prompt}} \text{ vs. Augmented Correctness } S_{\text{aug}}) \quad (29)$$

In contrast, user prompt word count demonstrated a weaker correlation with baseline real-image correctness ($r = 0.3107$) and an inverse correlation with output token length ($r = -0.1367$). This structural dynamic highlights two key mechanisms governing language-vision interactions in embodied robotics:

1. **Attentional Alignment in Detailed Queries:** Short, unconstrained user prompts (e.g., “Describe the scene”) allow the VLM’s cross-attention mechanism to wander across visual feature maps. When visual overlays (such as bounding boxes or segmentation masks) are introduced to short prompts, models often experience perceptual confusion. Conversely, detailed multi-clause prompts (e.g., “Locate the red fire extinguisher relative to the door frame and verify if the path is clear”) provide explicit semantic constraints that align with visual prompt overlays, directing cross-attention toward the target ROI and yielding higher factual correctness (S_{eval}).
2. **Mitigating Synthetic Visual Overlay Ambiguity:** When visual prompt transformations introduce artificial artifacts (such as synthetic bounding lines, HUD labels, or depth color maps), brief queries occasionally cause decoders to mistake these overlays for physical objects in the environment. Detailed task descriptions explicitly instruct the language decoder to interpret synthetic overlays as structural guidance rather than environmental clutter, effectively suppressing text-induced visual hallucinations.

6.4 Temporal Video Keyframe Dynamics and Edge Processing Trade-offs

Benchmarking 20 keyframe extraction algorithms across continuous mobile robotics video sequences highlights three structural insights regarding temporal prompt engineering:

6.4.1 The Failure of Traditional Shot-Boundary Detection in Continuous Motion

Standard computer vision video summarization algorithms (such as `scenedetect`) rely on sharp color histogram changes to identify shot cuts. While effective for edited cinema or broadcast media, this assumption breaks down in mobile robotics. Continuous camera translation and smooth pan-tilt movements do not produce discrete scene transitions, causing threshold-based algorithms to detect zero cuts in 94.29% of operational sequences. This demonstrates that mobile robotics video requires continuous motion-variance tracking rather than discrete transition detection.

6.4.2 The Recall vs. Token Budget Trade-off in Downstream VLMs

Video-based visual prompting introduces a strict trade-off between trigger frame recall, pre-processing latency, and autoregressive decoding cost. Fixed-capacity methods (e.g., `kmeans`, `laplacian`) that saturate the full 10-frame budget eliminate undersampling (0.00%), but force the downstream VLM to process an average of +8.59 redundant visual frames per query.

Our proposed `smart_sampling` strategy reduces the frame budget to 3.50 ± 3.82 frames on average, cutting visual token processing overhead by over 65% while achieving the highest exact-match rate (42.86%). In contrast, `qa_iframes` achieves the tightest semantic pruning by filtering candidate *I*-frames down to just 1.74 ± 2.33 frames (+0.33 extra frames), though its strict CLIPSeg presence threshold ($\tau_{\text{presence}} = 120$) leads to a 48.57% undersampling rate under degraded lighting conditions. For robotic applications requiring a deterministic zero-risk capture guarantee with minimal compute, native codec demuxing (`iframes`) provides a 0.00% undersampling rate at sub-second extraction latencies (0.45 ± 0.40 s).

6.4.3 Query-Conditioned Dynamic Frame Budgeting

A fundamental limitation of classical video summarization is its reliance on purely visual heuristics (e.g., optical flow magnitude, SSIM inter-frame divergence, or pixel gradients) to select keyframes. In embodied perception, visual motion does not inherently indicate task relevance: high visual variation (such as navigating down a plain hallway) may require only a single temporal snapshot, whereas subtle, low-motion manipulation events often require multiple fine-grained temporal steps. Developing end-to-end, lightweight keyframe selectors that dynamically allocate the frame budget based on the semantic complexity of the natural language query scaling from single-frame budgets for direct target identification up to multi-frame sequences for compositional reasoning is essential to minimizing edge token bloat without sacrificing task correctness.

6.5 Functional Classification and Taxonomy of Visual Prompt Families

To systematically evaluate the 35 visual prompt transformations benchmarked across Gemma-4 (12B), LLaVA-Phi3 (3.8B), and Qwen3-VL (4B), we group these strategies into five core algorithmic families based on their underlying feature modification mechanism. Table 10 outlines this taxonomy, detailing the computational pre-processing overhead, primary target tasks, and model architectural suitability.

Algorithmic Family Dynamics: As detailed in Table 10, the trade-off between pre-processing delay and VLM accuracy depends heavily on the chosen strategy family:

- **Photometric Regularizers** introduce minimal compute overhead ($\leq +1.3$ s), making them ideal for high-frequency low-light navigation.
- **3D Monocular Depth Maps** provide explicit geometric cues that boost spatial reasoning performance in larger models like Gemma-4 (12B), but incur heavy pre-processing latencies (+8.6 s for ZoeDepth).

Table 10: Functional Taxonomy of the 35 Visual Prompt Transformations Grouped by Algorithmic Family.

Algorithmic Family	Representative Methods	Pre-Processing	Target Task Domain	Architectural Suitability
Photometric Regularization	Gamma Correction, CLAHE, Color Jittering, Solarization, RetinexSSR	Low ($\leq +1.3$ s)	Low-Light Navigation, Hazard Detection	Highly synergistic with compact models (LLaVA-Phi3) by stabilizing specular contrast.
Spatial Focus & Cropping	Context-Aware Zoom (CAZ), Saliency Crop, Perspective Transform	Moderate (+1.5 s)	Target Disambiguation, Small Object Bounding	Effective for compact encoders; increases effective token density on ROI.
3D Monocular Depth Maps	ZoeDepth, MIDAS, DepthAnything, Surface Normalization	High (+4.2–+8.6 s)	Spatial Reasoning, Depth Ordering, Layout Analysis	Highly effective on large encoders (Gemma-4 12B); resolves 3D spatial ambiguity.
Bounding & Annotation Overlays	YOLO-World Annotation, QABB, Pose Estimation, RT-DETR	Moderate (+1.8 s)	Object Localization, Human-Robot Interaction (HRI)	Excellent for high-capacity models (Qwen3-VL); provides crisp spatial anchors.
Structural Masking & Removal	FastSAM, MobileSAM, Rembg, GridMask Dropout, Noise Injection	High (+4.7–+9.1 s)	Background Removal, Dense Clutter Isolation	Regularizes attention on noisy backgrounds, though dense masks can obscure fine textures.

- **Structural Masking** prevents attention collapse on cluttered backgrounds, but semi-transparent color masks risk covering fine surface details if segmentation backbones are over-saturated.

6.6 Qualitative Failure Taxonomy

Systematic examination of incorrect predictions identifies three primary structural failure modes associated with synthetic visual prompts:

1. **Mask Occlusion from Dense Segmentation:** High-density segmentation backbones (*FastSAM*, *MobileSAM*) render semi-transparent color masks over target boundaries. On compact models such as LLaVA-Phi3 (3.8B), these overlays occasionally obscured subtle surface textures (e.g., text labels, small surface fractures, or fine structural edges), leading to localized feature misclassification.
2. **Visual Clutter in Text-Based Graph Overlays:** Overlaying structured text nodes and relational edges directly onto 2D images (*SGG*) yielded lower correctness scores across all evaluated models (58.63% Gemma-4, 61.91% LLaVA-Phi3, 74.85% Qwen3-VL). Vision encoders frequently conflated synthetic text elements with physical objects in the primary scene.
3. **Anchor Loss in ROI Cropping:** Although *Context-Aware Zoom (CAZ)* performed effectively in targeted queries, cropping isolated regions of interest occasionally removed surrounding spatial reference frames. In *Spatial Reasoning & Layout Analysis*, isolating target objects impaired the model’s ability to determine relative distances and layout orientation.

6.7 Operational Deployment Framework

Based on the empirical findings across 12,250 evaluation instances, we synthesize a four-part deployment guideline tailored for edge-constrained mobile robotics:

Guideline 1 (Real-Time Closed-Loop Control Budgets): For time-critical mobile operations (> 0.2 Hz), deploy LLaVA-Phi3 (3.8B) paired with lightweight prompt transformations such as *Context-Aware Zoom (CAZ)* (4.47s latency) or *Query-Aware Bounding Box (QABB)* (4.96s latency). These proposed spatial prompts maintain -5.0s inference cycles while stabilizing localized feature extraction on compact edge decoders.

Guideline 2 (Cluttered Environment Manipulation): For fine-grained object manipulation and target disambiguation in dense clutter, deploy Gemma-4 (12B) combined with *Query-Aware Bounding Box (QABB)* or foreground depth isolation (*ZoeDepth*). Explicitly anchoring cross-attention with bounding overlays improves

localized object detection accuracy by up to +8.81% while pruning unneeded output tokens.

Guideline 3 (Low-Illumination Navigation): Apply photometric contrast pre-processing (*CLAHE* or *Gamma Correction*) prior to VLM tokenization in low-light conditions to stabilize hazard detection performance.

Guideline 4 (Offline Auditing and Supervision): Reserve high-capacity models such as Qwen3-VL for offline supervisory analysis. While achieving high accuracy on multi-hop reasoning (91.62%), its processing latency (54.25 s) and thermal load constrain its use in high-frequency onboard control loops.

6.8 Limitations and Operational Constraints

Several methodological, hardware, and software implementation constraints bound the findings of this study:

1. **Human-in-the-Loop Temporal Ground-Truth Verification:** In continuous robotics video curation, verifying whether candidate keyframe subsets successfully captured transient, short-duration ROI events (e.g., momentary gesture cues or fleeting dynamic obstacle entrances) remains an intensive manual process. Because existing evaluation metrics lack reliable zero-shot factual grounding mechanisms without heavy frame-by-frame VLM decoding, human auditing remains necessary to establish ground-truth trigger validity across multi-modal video sequences.
2. **Temporal Redundancy and Closed-Loop Video Extraction Latency:** Standard automated sampling heuristics (such as fixed-interval 1 Hz extraction) introduce substantial temporal redundancy, forcing vision encoders to process uninformative background frames prior to encountering the critical trigger event. While adaptive extractors (such as `smart_sampling` and `qa_iframes`) mitigate token bloat, the computational latency required to semantically filter candidate frames on edge hardware currently bounds high-frequency closed-loop video deployment.
3. **Hardware Thermal Throttling and Compute Overhead:** Evaluating large vision-language models like Qwen3-VL on edge hardware (NVIDIA Jetson AGX Thor) induces significant thermal load. Sustained tensor processing over extended evaluation passes triggered system thermal management limits, constraining high-parameter architectures to single-pass evaluation protocols ($N_{\text{run}} = 1$). Additionally, multi-stage computer vision pipelines (such as *Perception Metric Grounding* and *ZoeDepth*) introduce up to +9.10s of pre-processing delay, limiting their applicability in high-speed dynamic tracking scenarios above 1 Hz.
4. **Software Infrastructure and Cross-Engine Streaming Instability:** During continuous long-duration benchmark evaluations, web-based streaming interfaces deployed on Gecko-based browsers (Mozilla Firefox) experienced asynchronous WebSocket connection drops and socket timeouts during heavy token generation passes. Conversely, Chromium-based execution environments demonstrated persistent session stability under identical network loads. This highlights a software engineering deployment constraint regarding browser background throttling and socket buffer management during long-running robotics monitoring tasks.
5. **Sensor Degradation in Complete Darkness:** Photometric enhancement relies on residual ambient light. In total darkness (0 lux), passive visual prompt transformations fail to extract meaningful image features, underscoring the necessity of active thermal or infrared sensing channels in unlit operational environments.
6. **Deterministic Decoding Variance Across Backends:** Methods like PySceneDetect exhibit minor frame-boundary shifts when decoded across different OpenCV/FFmpeg backends, requiring explicit parameter clamping.
7. **Uniform Frame Budgets** fixed frame budget causes redundant oversampling in short clips and temporal under-sampling in longer ones, yet dynamically finding the optimal frame limit introduces its own search and latency trade-offs prior to inference.

7 Conclusion and Future Work

This thesis presented an empirical evaluation of 35 spatial visual prompt transformations and 20 dynamic keyframe extraction strategies across VLMs deployed on mobile robotics edge hardware (NVIDIA Jetson AGX Thor). Across 12,250 multimodal benchmark instances and continuous robotic video sequences, we systematically evaluated baseline versus augmented semantic correctness (S_{base} vs. S_{aug}), on-device execution latencies (t_{orig} vs. t_{proc}), semantic alignment (BERTScore, BLEU, CLIPScore), and downstream temporal reasoning fidelity.

7.1 Key Empirical Summary

Our benchmark findings demonstrate critical trade-offs across model scales, spatial prompt overlays, and temporal video extraction pipelines:

Spatial Visual Prompting Insights:

- 1. Receptive Field Focusing:** Visual prompting acts as an attentional spotlight. In *Object Detection & Localization*, spatial bounding and zoom overlays yielded a +8.81% accuracy increase for Gemma-4 (12B). However, this localized focus incurs a recall penalty in broad tasks like *Contextual & Scene Understanding* (−19.53%).
- 2. Real-Time Edge Execution Budgets:** Compact models such as LLaVA-Phi3 (3.8B) achieve fast onboard execution (5.08s per query, $\Delta t = +1.74\text{s}$ overhead) with tight token stability, offering a critical latency advantage over larger architectures for real-time mobile control loops.
- 3. Low-Light Photometric Resilience:** Photometric contrast filters (*CLAHE*, *Gamma Correction*) effectively expose object borders in dark operational scenes, increasing low-light hazard detection accuracy across all architectures.
- 4. Offline Supervisory Ceiling:** High-capacity architectures such as Qwen3-VL establish the highest overall accuracy ceiling (91.62% in multi-hop reasoning), but their heavy compute latency (54.25s) and thermal load restrict their deployment to offline supervisory auditing rather than onboard closed-loop control.
- 5. Prompt Complexity and Overlay Synergy ($r = 0.40\text{--}0.45$):** Across all three model families, user prompt length exhibited a strong positive correlation with augmented correctness. Detailed, multi-clause queries provide the semantic constraints necessary for decoders to interpret synthetic visual prompts as structural guidance, whereas short, unconstrained prompts (e.g., “describe the scene”) frequently induce perceptual confusion and overlay hallucinations.
- 6. Semantic Divergence and Automated Metric Blindspots:** Evaluating linguistic embedding similarity alongside grounded correctness revealed an automated metric decoupling. Structural overlays like *SGG* retained high semantic alignment (BERTScore $\approx 0.78\text{--}0.80$), yet suffered severe factual correctness penalties (58.84% on Gemma-4 and 57.87% on LLaVA-Phi3). This demonstrates that text similarity metrics alone cannot reliably evaluate embodied perception without grounded, multi-agent factual auditing.

Temporal Keyframe Extraction Insights:

- 1. Static Frame Allocation Inefficiency:** Fixed-budget keyframe methods consistently saturate upper frame thresholds (averaging +8.59 redundant frames over ground-truth requirements), introducing substantial visual token bloat without contributing new semantic information.

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2. **Variable Sampling and Semantic Pruning Trade-Offs:** Adaptive motion-aware strategies (`smart_sampling`, `tsdpc`) balance redundancy (averaging 2.09–3.66 extra frames) with strong task recall. Meanwhile, semantic query-guided demuxing (`qa_iframes`) achieves maximal token compression (1.74 ± 2.33 frames, +0.33 extra frames), establishing a Pareto frontier between token economy and spatial presence filtering.
 3. **Latency–Selectivity Dichotomy:** Native codec parsing (`iframes`) offers the lowest latency footprint (0.45 ± 0.40 s) with zero under-sampling failures (0.00%), whereas heavy perceptual pipelines (`katna`) introduce massive runtime overhead (132.64 ± 661.80 s) that hinders real-time edge deployment.
 4. **Failure Modes of Isolated Boundary Detectors:** Relying strictly on hard scene cut detection (`scenedetect`) leads to catastrophic under-sampling (94.29% failure rate) in continuous robotic video feeds, starving downstream models of essential visual context.

7.2 Future Research Directions

Building upon the empirical insights, we outline six key directions for embodied VLM perception and video understanding:

1. **Autonomous Meta-Policy Selector for Multi-Modal Pre-Processing:** Developing a lightweight, learned meta-controller that analyzes input query semantics and environmental conditions to dynamically select the optimal visual prompt transformation for static images (e.g., *QABB*, *CAZ*, or *CLAHE*) and the best-suited keyframe extractor for video streams (e.g., selecting `iframes` for zero-drop real-time navigation vs. `qa_iframes` / `smart_sampling` for token-efficient targeted reasoning).
2. **Query-Aware Semantic Keyframe Selection with Dynamic Thresholding:** Advancing semantic keyframe front-ends (such as `qa_iframes`) with adaptive, entropy-conditioned activation thresholds to eliminate the 48.57% undersampling penalty in low-contrast scenes without expanding the frame budget.
3. **Hardware-Accelerated In-Codec Extraction:** Offloading feature divergence calculations and adaptive frame sampling directly to GPU-accelerated video decoders (e.g., NVDEC or PyAV pipelines) to eliminate edge pre-processing latency penalties.
4. **Adaptive Dynamic Visual Prompting:** Designing real-time gating mechanisms that dynamically toggle visual prompt overlays based on query type, applying bounding crops for localization tasks while suppressing overlays during open-ended scene mapping.
5. **Multi-Modal Active Sensing Integration:** Integrating active thermal and infrared modalities into visual prompting pipelines, restoring spatial layout perception under total ambient darkness (0 lux).
6. **Edge Hardware Co-Design for Dynamic-Resolution Encoders:** Optimizing vision transformer attention kernels on edge hardware to process high-frequency synthetic visual overlays without triggering exponential token expansion delays.

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